

Valuation in a Private Equity Setting

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Chapter Overview

This chapter takes the perspective of a private equity investor and applies the hybrid valuation methods, which are particularly well suited to the types of investments made by private equity firms. These investment companies raise funds in limited partnerships with a relatively short life span of 7 to 10 years. They then invest in companies that span all stages of the life cycle of the firm—from venture capital to vulture capital. In all cases, however, their investments are of relatively short duration, with targeted investment exit in 4 to 6 years. The short duration magnifies the importance of the terminal-value component of the valuation, and because the exit often involves the public capital markets, market-based multiples are commonly used.

We consider two specific private equity valuation examples in this chapter. The first is the deal-structuring problem that arises when a venture capital firm makes an investment in a start-up firm. The second example is the valuation of a build-up leveraged buyout transaction by an LBO fund. In both cases private equity capital is the common denominator, although the purpose of the investment (providing start-up or growth versus restructuring capital) is quite different.

1 INTRODUCTION

Perhaps the most visible difference between how the valuation problems were approached earlier and the way in which private equity firms evaluate investments relates to the discount rate. Up to this point, our valuation problems used discount rates in the 8% to 12% range. However, most private equity investors claim to require an internal rate of return in the 25% to 50% range and even higher for start-up investments. How do they justify such high required rates of return? There are four possible avenues that can be taken in this regard. The first avenue relates to the fact that these investments are quite risky. They involve risky investment propositions related to the financing of start-up

¹ We would like to express our gratitude to J. William Petty for his many helpful insights in the preparation of this chapter.

firms, the provision of growth capital, or the restructuring of older firms. In the case of buyout firms, they entail the use of substantial amounts of financial leverage. Moreover, investments in private equity funds do not benefit from the liquidity offered by public investments in stocks, bonds, or mutual funds, and thus require a return premium over these alternative investments.

A second justification for the very high required rates of return is that these rates of return include compensation to the private equity firm for more than just the use of money. They also incorporate the value of the expertise that the private equity firm partners bring to the deal. The partners generally take a seat on the board of the firms in which they invest and provide advice to start-ups and actually control buyout investments.

The third justification for the very high required rates of return of private equity firms is that they generally evaluate what we refer to as “hoped-for” cash flows rather than expected cash flows. Consequently, we might interpret the 25% to 50% required rates of return as hoped-for returns rather than expected rates of return.

Finally, it should be emphasized that the required rate of return on these investments is determined by the opportunity cost of capital, which means that when private equity firms have an abundance of good opportunities and scarce capital, they can require very high rates of return. However, when there is abundant capital and fewer opportunities, we expect required rates of return to decline.

There are some unique features of how the hybrid valuation approach is applied by private equity firms that we describe in this chapter. First, in most cases when a private equity firm makes an investment, it does not expect to receive significant cash flows in the initial planning period, so the valuation emphasis is on the (terminal) value of the enterprise at a future date when the private equity investor hopes to cash out of the investment. This value is generally estimated as a multiple of EBITDA. Second, the object of the valuation exercise generally focuses on the value of the investors’ equity investment, rather than on the enterprise value of the company receiving funding. Neither of these features provides significant difficulties for the hybrid valuation approach, but they do give rise to some unique terminology and special adaptations of various tools.

Up to this point, we have focused our valuation approach on situations where we evaluate whether to invest a specified amount of cash in an investment opportunity that offers an uncertain set of future cash flows. From the perspective of the venture capitalist (VC), the investment problem is similar in that start-up firms typically require a specific amount of external funding in order to produce an uncertain set of future cash flows. However, the VC’s problem is different in that he or she must not only evaluate the uncertain future cash flows but also negotiate the ownership fraction of the start-up’s shares that he or she must have in order to earn a desired return on the investment. Thus, the valuation problem for the VC is generally discussed in the context of deal structuring. Although this is a somewhat different problem from those we have addressed so far, we can solve it with a slight modification of our valuation methods.

This chapter is organized as follows: In Section 2, we provide a brief overview of the private equity and venture capital markets. Section 3 provides an overview of deal-structuring analysis as it is carried out by venture capital firms. In Section 4, we

Did you know?

How significant is the market for private equity?

The largest private equity firms now raise multibillion-dollar funds, so by participating in groups or clubs, they are able to garner the billions needed to acquire some of the largest publicly held firms. For example, on September 15, 2006, four private equity firms (Blackstone Group Funds, Texas Pacific Group, Permira Group Funds, and the Carlyle Group) agreed to pay \$17.6 billion for Freescale Semiconductor Inc. (Motorola’s spin-off of its semiconductor operations). However, the downturn in the economy that began in 2006 depressed private equity fund investing for two years. By 2009, private equity funds were holding almost \$400 billion in unspent funds.

INDUSTRY INSIGHT

Private Versus Public Equity Investing

Private equity investing differs in four ways from public equity investing:

1. **Private equity investors make *illiquid* investments that cannot be sold because of either a lack of an organized market or investment restrictions.** The investment might take the form of equity or debt securities issued by a privately held firm or restricted shares of stock issued by a publicly held firm.² In either case, the investment cannot be easily sold. In contrast, mutual funds and other passive investors hold publicly traded debt and equity securities that can be bought or sold at any time.
2. **Private equity investors are *active* investors.** They generally take an active role in managing the companies in which they invest (for example, taking a seat on the board) or serve as financial advisors to the firm's management. This contrasts with the "hands-off" investment style followed by mutual funds and others who invest primarily in the public equity markets.
3. **Private equity investments are made for a *finite* period of time.** Traditional private equity investing involves ownership of limited partnership units that are part of a private equity fund. The partnership typically has a fixed life of 7 to 10 years, at which time the fund is liquidated and the proceeds are distributed to the partners. Mutual funds and other entities that invest in public equity have no set liquidation date and correspondingly can have a much longer investment horizon.
4. **Private equity investments are risky and illiquid and thus require high returns.** Because the failure rate can be quite high in some types of private equity investing and the investment is illiquid for years, private equity investors typically demand very high promised rates of return.

look at the valuation of an investment by an LBO firm. We close this section with an application of the APV model for a private equity investment. Finally, Section 5 contains summary comments.

2 OVERVIEW OF THE MARKET FOR PRIVATE EQUITY

Before we launch our discussion of the valuation of private equity investments, it will be helpful to describe the market for private equity and to gain some perspective on the role of private equity finance in the U.S. economy. When we refer to a private equity firm, we are referring to a financial intermediary that is in the business of raising pools of capital (generally in limited partnerships) and using that capital to invest in companies that need financing. Essentially, private equity is an ownership stake either in a

² The restriction we refer to relates to the fact that the investor cannot sell his or her shares for a prescribed period of time.

private company or in shares of a public company that are restricted such that they cannot be sold for some specified period of time.³ In general, a private equity investor is an active investor who acquires some measure of control over the firms in which he or she invests, often by serving on the firm's board. The Industry Insight box on the opposite page summarizes the differences between public equity and private equity investing.

Market for Private Equity—Financial Intermediaries

Figure 1 summarizes the three parties involved in the private equity market. These include suppliers of funds (e.g., the investors in the private equity partnerships), the private equity investment companies that invest in and manage these funds, and the businesses that use private equity funds. Although it is difficult to know exactly how large the various components of the market are at any moment, it is generally believed that approximately 80% of the funds that flow into private equity come from public and private pension funds. The majority of these funds are then invested by the private equity firms in nonventure activities, which include meeting the expansion, recapitalization, and reorganization needs of companies, as well as acquiring large public firms (see the Industry Insight box on “Biggest Buyout Deals of 2006” two pages ahead).

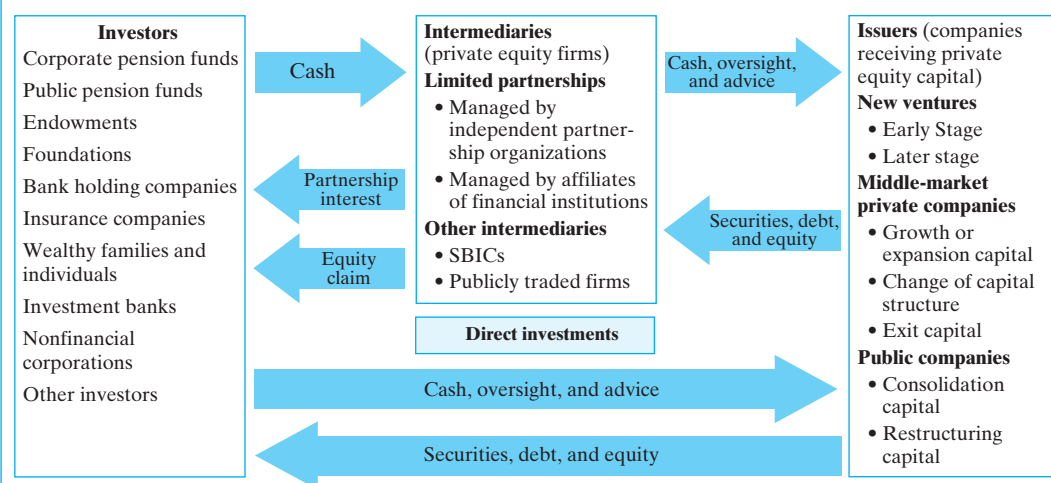
Investors—The Suppliers of Private Equity Finance

Most private equity firms are organized as *limited partnerships* that raise capital from a small number of sophisticated investors in a private placement. The investors in the fund include public and private employee pension funds, university endowment funds, wealthy families, bank holding companies, and insurance companies. The largest segment (as we noted above) is public and private pension funds, which contribute approximately 80% of all the funds in the private equity market.

Investments—The Demand for Private Equity Finance

Although private equity funds invest in the full spectrum of businesses, individual private equity firms tend to specialize in a very narrow range of activities that coincide roughly with the life cycle of the firms in which they invest, as depicted in Figure 2. Seed capital for start-ups is typically supplied by wealthy individuals commonly referred to as “angel investors” or “business angels.” This type of investing typically does not involve the use of an intermediary (i.e., it is a form of direct investing, as depicted at the bottom of Figure 1). Venture capitalists typically provide established start-up companies with early-stage financing that can carry the companies up to the point where they need to access the public capital markets for their funding needs or where they are sold to other companies (often public companies). This is typically the earliest stage in the firm's life cycle in which a venture capital firm will invest. See the Practitioner Insight box for a discussion of the thought process that goes into determining the size of a VC fund and the minimum level of investment in a portfolio company.

³ These investments are referred to as private investments in public equities or PIPEs.

Figure 1 The Market for Private Equity**Private Equity Firms:**

Limited Partnership—The standard vehicle for raising funds to invest in private equity is a limited partnership that typically has a fixed life of 10 years. The partnership's general partner makes investments, monitors them, and finally exits to earn a return on behalf of the investors—limited partners.

SBIC—Small business investment corporations were created in 1958. They are privately managed, for-profit investment funds formed to provide equity and/or debt capital to U.S. small businesses.

Categories of Private Equity Investments:

Seed and start-up capital—Typically provided by angel investors (i.e., wealthy individuals), this type of capital is provided to develop a concept, create the initial product, and carry out the first marketing efforts. The company that seeks financing is typically very young (less than one year old) and has not produced a product or service for commercial sale.

First-stage capital—The money provided to an entrepreneur who has a proven product, to start commercial production and marketing. This is typically the earliest stage at which a venture capital firm will invest.

Second-stage capital—The capital provided to expand marketing and meet the growing working capital needs of an enterprise that has commenced production but does not have sufficient cash flows from its operations to fund its capital needs.

Expansion capital—The capital needed to fund the expansion of a profitable firm's operations where the growing firm is incapable of generating sufficient earnings internally to fund its needs for capital.

Bridge capital—A short-term loan that provides needed capital while the borrower arranges for more comprehensive, longer-term financing.

Mezzanine capital—Unsecured, high-yield, subordinated debt or preferred stock that represents a claim on a company's assets that is senior only to that of a company's shareholders. The term *mezzanine* comes from the fact that these forms of financing are somewhere between debt and equity in terms of the priority of their claim on the firm's earnings and assets in the event of default or bankruptcy.

Source: Based on Figure 3 in Stephen D. Prowse, "The Economics of the Private Equity Market," *Economic Review of the Federal Reserve Bank of Dallas*, 3rd Quarter, 1998, 21–34.

Biggest Buyout Deals of 2006

By far the largest fraction of all private equity investing involves nonventure, buyout financing. The following list contains the five largest buyout deals for 2006, which was the largest dollar-volume year of buyouts in U.S. history. Note that with only one exception, the investments involve multiple private equity firms. The reason for this relates to the magnitude of the investment being made and concerns by the private equity investors that they not let their investments become too concentrated in a small number of very large investments.

Date	Acquisition Target	Private Equity Buyer(s)*	Equity Value
May 29, 2006	Kinder Morgan Inc.	GS Capital Partners LP, AIG, Carlyle Group LLC, and Riverstone Holdings LLC	\$27.4 billion
July 24, 2006	HCA Inc.	Bain Capital LLC, Kohlberg Kravis Roberts & Co., Merrill Lynch Global Private Equity Group, and Riverstone Holdings LLC	\$32 billion
October 2, 2006	Harrah's Entertainment Inc.	Apollo Management LP and Texas Pacific Group Inc.	\$25.8 billion
November 15, 2006	Clear Channel Communications Inc.	Bain Capital Partners LLC and Thomas H. Lee Partners LP	\$26.8 billion
November 19, 2006	Equity Office Properties Trust	Blackstone Group	\$32.5 billion

***Descriptions of the Private Equity Firms (from their Web sites):**

AIG Private Equity Ltd—A Swiss investment company that invests in a diversified portfolio of private equity funds and privately held operating companies.

Bain Capital LLC—Established in 1984, Bain Capital has approximately \$40 billion in assets under management. Bain Capital's family of funds includes private equity, venture capital, public equity, and leveraged debt assets.

Blackstone Group—Formed in 1987, the firm managed \$28 billion through its Blackstone Capital Partners I, II, III, IV, and V and Blackstone Communications Partners funds.

Carlyle Group LLC—Formed in 1987, the Carlyle Group is one of the world's largest private equity firms, with more than \$46.9 billion under management with 46 funds, across four investment disciplines including buyouts, venture and growth capital, real estate, and leveraged finance.

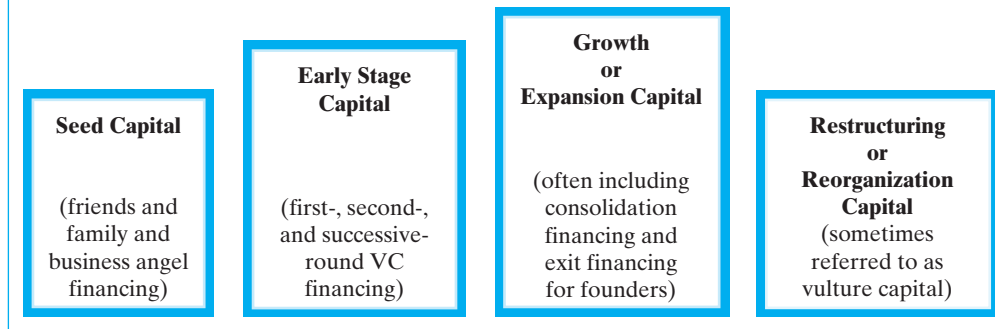
GS Capital Partners LP—Makes private equity investments on behalf of Goldman Sachs and others. The firm's current fund, GS Capital Partners V, is an \$8.5 billion fund, with about \$2.5 billion committed by Goldman Sachs and its employees, and the remainder committed by institutional and individual investors. The firm invests in a range of industries in a variety of situations, including build-ups, leveraged buyouts, recapitalizations, acquisitions, and expansions.

Kohlberg Kravis Roberts & Co.—Established in 1976, KKR has completed over 140 transactions valued at approximately \$226 billion. As of September 30, 2006, these transactions were valued at approximately \$70 billion on \$27 billion of invested capital. KKR's investors include corporate and public pension plans, financial institutions, insurance companies, and university endowments.

Merrill Lynch Global Private Equity Group—The private equity investment arm of Merrill Lynch provides capital to fund growth, financial restructuring, or change of control. MLGPE typically holds investments for the medium-to-long term (three to seven years).

Riverstone Holdings LLC—A \$6 billion private investment firm founded in 2000. The firm has completed more than 20 transactions and is one of the largest private equity investors in the energy and power industry.

** Unless otherwise noted, these data are for 2006.

Figure 2**Stages in the Life Cycle of the Firm**

Growth and expansion capital is provided by a mixture of private equity firms, including some firms that consider themselves venture capitalists and others that fall into the buyout category. Finally, providing restructuring or reorganization capital is the domain of the buyout or LBO firm. The motive of the buyout firm can be to finance a reorganization that rationalizes and reconfigures the firm to become competitive or to break up and dissolve the firm's operations. In the latter case, the buyout firm is euphemistically referred to as a "vulture fund."

3 VALUING INVESTMENTS IN START-UPS AND DEAL STRUCTURING

When a successful new company grows at a pace that cannot be financed by the entrepreneur alone, the entrepreneur frequently seeks funding from a venture capital firm in exchange for partial ownership of the firm. In the negotiation process, a critical issue for the entrepreneur is: How much of the company's equity will the outside investor demand in exchange for providing the needed capital? The amount that the entrepreneur must give up depends on the VC's assessment of the firm's value. This valuation, in turn, reflects the VC's evaluation of the firm's future prospects and the VC's target rate of return for the investment. Although the venture capital evaluation approach may appear unique, it is actually a variant of the hybrid valuation approach, which combines the methods of multiples and discounted cash flow.

The Cost of Capital for Venture Capital Financing

The venture capital evaluation approach, like any discounted cash flow (DCF) valuation method, requires two primary inputs: a discount rate and an estimate of future cash flows. Figure 3 reviews the two basic approaches that we have used to match cash flows and discount rates. The first is the traditional DCF approach to valuation, which discounts *expected* future cash flows using the *expected* rate of return from a

PRACTITIONER INSIGHT

Determining the “Size” of VC Fund to Raise—An Interview with Venture Capitalist Joe Cunningham, MD*

Historically, venture capital (VC) funds that invest in early-stage companies and start-ups have been relatively small when compared with buyout funds that buy mature businesses. However, in recent years, venture capitalists have raised some very large funds. For example, Austin Ventures, located in Austin, Texas, has over \$3 billion under management in nine funds with over a billion dollars raised in its fund VII, and Oak Investment Partners raised a \$2.56 billion venture fund in 2006. However, the typical venture fund remains small in comparison with the gargantuan buyout funds raised in recent years.

To determine the size of the venture fund that we needed to raise, my two partners and I approached the problem by asking some basic questions and letting the answers guide our choice of a fund size:

1. *How large a portfolio of companies can we effectively manage?*** Based on our experiences in founding and advising start-up companies in our target industry (medical technology and services), we estimate that we can each look after 5 companies, for a total of 15.
2. *What is the nature of the typical exit and the value at exit for a start-up company in our industry space?* To address this question, we studied the exits of start-ups in the Medical Technology and Services industry for the last decade. We found that approximately 85% of these exits involved the sale of the start-ups to larger companies, for a price between \$100 million to \$200 million with only a half-dozen IPOs. We concluded that we should make investments in firms that could be sold at our exit in the \$100–\$200 million range.
3. *What target return multiples (exit value/invested capital) should the fund use?* Given the risks and opportunities we saw in the industry, we decided on a target exit multiple of 10 times our invested capital. Of course, we realize that

this multiple will be reached only for the most successful investments and the average for the fund will probably be around 5 times. Nonetheless, our target when making an investment is 10.

4. *How much money should we plan to invest in each start-up?* To get to a \$100–\$200 million exit value, with an exit value/invested capital ratio of 10 times, requires post-money valuations*** for the start-ups at the time we invest that are in the \$10–\$20 million range. Thus, if we invest \$5 million in a start-up in return for 50% ownership of the firm, this is consistent with a post-money valuation of the firm’s equity of \$10 million. Using the 10-times target exit multiple implies a valuation of the start-up’s equity at exit of \$100 million. Similarly, if we invest \$10 million in the start-up company in return for half its equity, the post-money value of the startup will be \$20 million and it will have an exit value of \$200 million. Consequently, we anticipate investing from \$5–\$10 million in each start-up in two financing rounds.

Putting the answers to these questions together and assuming that we invest an average of \$8 million in each of 15 companies, we determined that the optimal fund size for us was \$120 million.

*Joe Cunningham, MD, is managing director for Santé Health Ventures, Austin, Texas.

**The answer to this question will vary from fund to fund, depending upon the industry in which the fund invests (which determines its capital intensity) and the type of value proposition the VC brings to its portfolio companies. Value proposition refers to the level of VC contribution; for instance, simple monetary investment, or additional specialized industry knowledge, and contacts that require more frequent involvement with the company.

***Post-money valuation is a commonly used term in the venture capital industry, and it refers to the value of the start-up business that is implied by the amount of money invested by the VC fund in combination with the fraction of the start-up company’s shares the VC acquires.

Figure 3**Alternative DCF Valuation Approaches**

Approach	Cash Flow	Discount Rate	Applications
Traditional DCF or expected-return valuation model	Expected cash flow	Expected rate of return on comparable-risk investment	Project and business valuation • Traditional WACC approach • Adjusted present value approach
Optimistic or “hoped-for” valuation approach	Optimistic cash flow	Optimistic or “hoped-for” rate of return	• Bond valuation • Venture capital valuation • LBO valuation

comparable-risk investment (i.e., the opportunity cost of capital). This DCF model is the de facto standard for most applications in finance and was the basis for most of our discussions in prior chapters. The second method is the optimistic or hoped-for DCF valuation approach.⁴ This variant of the DCF model discounts optimistic or hoped-for cash flows back to the present using optimistic or hoped-for rates of return. We believe that the venture capital method falls into this latter category, although this terminology is not used by venture capitalists.

In general, the hoped-for cash flows used to value start-up firms are those that materialize in scenarios in which everything goes as planned. The optimism embedded in these forecasts comes from the fact that entrepreneurs (who are generally very optimistic about the prospects for their businesses) are the primary source of these cash flow estimates. Venture capitalists like to encourage the enthusiasm of the entrepreneurs they fund, so rather than trim optimistic projections, they generally require rates of return that are high enough to offset the entrepreneurs' optimistic cash flow estimates, as well as compensate the venture capitalist for the risk of the venture. Besides, the venture capitalists can turn the entrepreneur's optimistic forecast into ambitious targets that the entrepreneur can be held to in the future.

Venture capital investor-required rates of return (ROR) vary with the stage of the investment. For example, in the earliest stage of investment, referred to as the *seed* or *start-up stage*, venture capitalists typically require rates of return ranging from 50% to 100% (per annum!). As the firm progresses through the start-up phase

⁴ There is yet a third variation of the DCF model that we did not include in Figure 3, and it is the certainty equivalent model. This model discounts *certainty equivalent* cash flows (i.e., risks adjusted to equal their certain equivalents) using the *risk-free* rate of interest. We will discuss this particular model later, when we consider futures and options.

**PRACTITIONER
INSIGHT**
**Investing in Private Equity—
An Interview with
Jonathan Hook***
How do you think about rates of return you target for private equity investing?

The lack of liquidity in private equity investments is a critical consideration for anyone who considers making this type of investment. Investing in a private equity fund generally means you are tying up your money for several years. Consequently, as a general rule of thumb, we require that private equity investments provide the opportunity to earn 4% to 5% more than the return we anticipate on the S&P 500.

How much of your endowment portfolio do you allocate to private equity?

A primary consideration for us is that we have a balanced and well-diversified portfolio, so we try not to get too overweighted in any one asset class. With private equity, we target 15% of our portfolio, but with the flexibility to move this allocation up or down by 5% as opportunities present themselves.

*Jonathan Hook is Vice President and Chief Investment Officer at Ohio State University Endowment Fund. This interview took place when Mr. Hook was the Chief Investment Officer at Baylor University.

into first- and second-round financings, the required rate of return drops, as the following table indicates.⁵

Stage of Investment ⁶	Annual ROR (%)	Typical Expected Holding Period (years)
Seed and start-up	50%–100% or more	More than 10
First stage	40–60	5–10
Second stage	30–40	4–7
Expansion	20–30	3–5
Bridge and mezzanine	20–30	1–3

Do venture capitalists actually realize these hoped-for returns? It is difficult to answer this question, because the private equity funds are just that, “private,” and do not report the performance of their funds on a regular basis. However, one recent study found that when fees are netted out, buyout funds earn slightly less than the average

⁵ Jeffrey Timmons and Stephen Spinelli, *New Venture Creation: Entrepreneurship for the 21st Century*, 6th ed. (Irwin, 2004).

⁶ See the legend to Figure 1 for definitions of each of the types of financing listed here.

return of the S&P 500 stock index.⁷ The point here is that the average rates of return realized on both venture and nonventure investments are much lower than the required rates of return that are used to discuss cash flows.

It is important to emphasize that these required rates of return reflect the opportunity cost of capital, which is in turn determined by the rate of return on alternative investments. See the Practitioner Insight box titled “Investing in Private Equity—An Interview with Jonathan Hook.” In the long run, we expect required rates of return to be determined by the alternate risk factors. However, in the short run, the required rates of return on venture capital and other private equity investments are determined by the supply and demand for capital in this sector. When there is lots of capital chasing very few deals, required rates of return are likely to be relatively low. On the other hand, when capital is tight in this sector or when there are lots of opportunities, required rates of return are likely to be higher.

We offer one final point regarding the high required rates of return for venture capital financing, which relates to the fact that the relationship between a start-up firm and its venture capitalist often goes beyond acquiring financing. Frequently, the start-up firm receives valuable business advice and business connections from the venture capitalist who sits on their board. Because VCs typically focus their investing narrowly in a limited number of industries, this expertise and knowledge of the industry can prove essential to the future success of the start-up. The compensation for this nonfinancial source of value is wrapped up in the cost of VC financing, which also helps explain why VC financing looks so expensive.

Valuing a VC Investment and Structuring the Deal

In most cases, a venture capitalist provides funding to an entrepreneur in exchange for stock in his or her business. The key variable that determines the deal structure is the valuation that the venture capitalist places on the entrepreneur’s business.

To illustrate how a venture capitalist values and structures a deal, consider the hypothetical case of Bear-Builders.com. The entrepreneur behind Bear-Builders.com started the business last year with an initial investment of \$1 million and now needs an additional \$2 million to finance the expansion of its already-profitable operations. To raise the funds, the entrepreneur approaches Longhorn Partners, a hypothetical venture capital firm located in Austin, Texas.

Investor Expectations

Longhorn believes that the opportunity is good and is willing to provide the \$2 million that Bear-Builders.com needs, in exchange for a share of the company’s common equity. Now the key question is: How much stock will the entrepreneur have to give up in the process of acquiring the needed funds? Bear-Builders.com is seeking first-stage financing, so Longhorn Partners’ required (i.e., hoped-for) rate of return is 50% per year. If Longhorn

⁷ See Steven Kaplan and Antoinette Schoar, “Private Equity Performance: Returns, Persistence and Capital Flows,” *Journal of Finance* 60, no. 4, pp. 1791–1823. Stephen D. Prowse, “The Economics of the Private Equity Market,” *Economic Review of the Federal Reserve Bank of Dallas*, Third Quarter (1998), 21–34, provides similar results for earlier time periods as follows:

Partnerships Formed in:	Venture Capital	Nonventure Capital	Public Small- Company Stocks
1969–79	23.3%	—	11.5%
1980–84	10.0%	24.8%	15.3%
1985–89	15.2%	15.3%	13.4%
1990–91	24.1%	28.9%	15.6%

BEHAVIORAL INSIGHT

Aligning Incentives Among the Participants in the Private Equity Market

Asymmetric information is the norm for business dealings. Very simply, the parties to a transaction do *not* share the same information, and this allows one or both parties to take actions that are personally beneficial but detrimental to the other.

Two sets of relationships are of concern in our discussion of private equity finance. The first deals with the relationship between the more-informed general partners, who can potentially take advantage of the less-informed limited partners. The

second relationship relates to the private equity partnership and the more-informed managers of the portfolio companies. The table below summarizes how partnerships try to limit the costs that arise from these information asymmetries by first performing extensive due diligence. They also control managerial behavior through the use of incentive-based contracts and by restricting access to partnership financing, because other financing sources are limited for the portfolio companies.

Means of Control	Agency Relationship	
	Limited Partners and General Partners in Private Equity Firms	Private Equity Firm and Portfolio Companies
Indirect: Performance Incentives (the “carrot”)	■ Reputation of general partner affects future fund-raising ability ■ General-partner compensation based on fund performance	■ Managerial ownership of company shares ensures a long-term interest in company performance ■ Managerial compensation tied to company performance
Direct: Contracts and Internal Controls (the “stick”)	■ Covenants of the partnership agreement place boundaries on what the general partner can do and provide for limited-partner intervention in extreme circumstances ■ Advisory boards (including limited-partner representation) provide direct oversight over general-partner decisions	■ The private equity firm generally takes a board seat and can gain added seats if performance of the portfolio company falters ■ The fact that the private equity firm controls the portfolio company’s access to additional financing provides a strong source of control over management of the portfolio company

has a five-year investment horizon⁸ and receives no cash distributions until the end of this period (which is frequently the case), then it needs to receive \$15,187,500 at the end of five years in order to realize a 50% per annum rate of return. That is,

$$\$2,000,000(1 + .50)^5 = \$15,187,500$$

Two variables determine whether the investor will actually receive \$15,187,500 in five years: (1) the value of the firm at the end of the five years and (2) the fraction of that value that belongs to the investor. Let’s consider each of these factors in turn.

Valuing Bear-Builders.com’s Equity

The VC’s investment return from providing start-up capital to Bear-Builders.com is derived from the value of Bear-Builders.com’s equity at the end of the VC’s planned investment

⁸ The term *investment horizon* refers to the desired holding period for an investment. Because VC firms raise their capital with limited partnerships that have a limited life (normally five to seven years), they typically have a very short investment horizon when they invest.

period (typically four to six years). The approach we take to value Bear-Builders.com's equity entails first estimating the firm's enterprise value and then deducting any interest-bearing debt the firm may have outstanding.

Estimating Enterprise Value at the End of the Planned Investment Period The first step in the valuation process is to come up with an estimate of enterprise value for Bear-Builders.com after five years, when the venture capitalist hopes to "harvest" or "exit" the investment by selling his or her shares. Venture capitalists typically calculate an estimate of the enterprise value of the firm on the harvest date by forecasting both the firm's ability to produce before-tax cash flows from operations, or EBITDA, and the appropriate multiple of EBITDA for valuing the firm. For example, if Bear-Builders.com's owners forecast the firm's EBITDA in five years to be \$6 million and the VC uses an EBITDA multiple of 5, then Bear-Builders.com will be worth \$30 million (\$6 million $\times$ 5) when Longhorn Partners plans to harvest its investment in the firm.

Note that there are typically no interim equity cash flows to the VC (or the entrepreneur) prior to the VC's exit. This results from the fact that all available cash is either reinvested in the firm or used to service the firm's debt.

To recap quickly, the calculations used to estimate the enterprise value of Bear-Builders.com at the end of five years are as follows:

EBITDA in Year 5	\$ 6,000,000
EBITDA multiple	$\times \quad 5$
Equals: Enterprise value	\$ 30,000,000

Calculating Equity Value at the End of the Planned Investment Period Now that we have estimated the firm's enterprise value at the end of five years (\$30,000,000), we can calculate the value of its equity as follows:

$$\text{Equity Value} = \text{Enterprise Value} - \text{Interest-Bearing Debt} + \text{Cash}$$

or, where

$$\text{Net Debt} = \text{Interest-Bearing Debt} - \text{Cash}$$

then

$$\text{Equity Value} = \text{Enterprise Value} - \text{Net Debt}$$

Note that we subtract only interest-bearing debt from enterprise value. Moreover, to estimate equity value, we typically add in the projected amount of cash in the firm on the harvest (i.e., exit) date to our estimate of equity value. This assumes that the firm's cash balance can be viewed as excess cash and therefore is not reflected in the EBITDA multiple used to estimate enterprise value. For example, if we expect Bear-Builders.com to have \$300,000 in cash at the end of five years, and if Bear-Builders.com is expected to have \$3 million in interest-bearing debt in five years, the company's equity value would be \$27.3 million, calculated as follows:

Enterprise value	\$30,000,000
Less: Net debt (i.e., interest-bearing debt less cash)	<u>(2,700,000)</u>
Equals: Equity value	\$27,300,000

Computing Ownership Interests—Defining the Deal Structure

Up to this point, we have shown that if all goes as planned, the value of Bear-Builders.com's stockholder equity in five years will be \$27.3 million. The venture capitalist requires \$15,187,500 at that time in order to earn a 50% per annum rate of return on its \$2,000,000 investment. Thus, to achieve this total return, the venture capitalist must own 56% of the firm's stock, computed as follows:

$$\begin{aligned}\text{Venture Capitalist's} \\ \text{Ownership Percentage} &= \frac{\text{VC's Required Equity Value (in Year 5)}}{\text{Estimated Equity Value for the Firm (in Year 5)}} \\ &= \frac{\$15,187,500}{\$27,300,000} = 56\%\end{aligned}$$

Under this deal structure, the founder will own the remaining 44% of the common stock ($100\% - 56\% = 44\%$), which would entitle him or her to shares worth \$12,112,500 in five years.

Summing Up the Venture Capital Method

We have briefly described the process used by venture capitalists to evaluate an entrepreneurial company and structure a financing agreement. The methodology combines the method of multiples based on comparable transactions along with discounted cash flow. Very simply, the investor (venture capitalist) determines the fraction of the firm's ownership it must get in order to satisfy its return requirement as follows:

1. The investor first determines a rate of return that he or she hopes to realize from the investment ($k_{\text{Hoped-for VC Return}}$).
2. The investor's required rate of return is used to determine the dollar value that the investor hopes to realize at the end of the planned holding period, H years (typically four to six), in order to justify making the initial investment in the firm plus cash less interest-bearing debt, i.e.,⁹

$$\left(\frac{\text{Venture Capital}}{\text{Investment}_{\text{Today}}} \right) (1 + k_{\text{Hoped-for VC Return}})^H = \left(\frac{\text{Required Value of the Venture Capital}}{\text{Investment}_H} \right) \quad (1)$$

⁹ This relationship holds only if the VC does not receive any interim cash flows, such as interest or preferred dividends in those cases where the firm issues debt or preferred stock. If interim cash flows are paid, then the right-hand side of Equation 1 is reduced by an amount equal to the future value of all such interim payments, compounded at the venture capitalist's hoped-for rate of return (e.g., if interim interest payments equal to Int_t are made each year up until Year H):

$$\begin{aligned}& \left(\frac{\text{Venture Capital}}{\text{Investment}_{\text{Today}}} \right) (1 + k_{\text{Hoped-for VC Return}})^H + \sum_{t=1}^H \text{Int}_t (1 + k_{\text{Hoped-for VC Return}})^{t-1} \\ &= \left(\frac{\text{Required Value of the Venture Capital}}{\text{Investment}_H} \right)\end{aligned}$$

3. The investor next estimates the value of the firm's equity at the end of the planned holding period using a multiple of the firm's projected EBITDA in Year H , i.e.,

$$\frac{\text{Estimated Equity}}{\text{Value}_H} = \text{EBITDA}_H \times \text{EBITDA Multiple}_H + \text{Cash}_H - \text{Debt}_H \quad (2)$$

4. Finally, the investor calculates the fraction of the firm's future value that will be needed to satisfy his or her total dollar-required return (i.e., the initial investment compounded at the investor's hoped-for rate of return over the investment period), as follows:

$$\frac{\text{Ownership}}{\text{Share}} = \frac{\left(\frac{\text{Required Value of the Venture Capital Investment}_H}{\text{Estimated Equity Value}_H} \right)}{\quad} \quad (3)$$

Thus, the VC method results in an estimate of the VC's ownership share that generates the appropriate hoped-for rate of return.

Pre- and Post-Money Value of the Firm's Equity

We have just pointed out that the venture capital approach focuses on the value of the firm at the future date when the venture capitalist plans to exit the investment. However, venture capitalists also estimate *the value today* for the businesses in which they invest. The implied value of the equity of the firm today is captured by something called the **post-money investment value**. For example, in the Bear-Builders.com example, the VC gets 56% ownership for an investment of \$2 million; this implies that the value of the firm's equity after (post) the VC firm's investment is \$3,571,428. That is,

$$\begin{aligned} \frac{\text{Post-Money Investment Value}}{\text{of the Firm's Equity}} &= \frac{\text{Funding Provided by the VC Firm}}{\text{VC Firm's Ownership Interest (\%)}} \\ &= \frac{\$2,000,000}{.56} = \$3,571,428 \end{aligned}$$

The post-money investment value is the focal point for negotiations between the entrepreneur and the VC. Basically, it provides a shorthand way to describe the structure of the VC deal.

Venture capitalists also use the term **pre-money investment value** to refer to the difference between the post-money implied value of the firm's equity and the amount of money that the venture capitalist puts in the firm. In this case, the pre-money investment value is \$1,571,428, i.e.,

$$\begin{aligned} \frac{\text{Pre-Money Investment Value}}{\text{of the Firm's Equity}} &= \frac{\text{Post-Money Investment Value}}{\text{of the Firm's Equity}} - \frac{\text{Funding Provided by the VC Firm}}{\quad} \\ &= \$3,571,428 - 2,000,000 = \$1,571,428 \end{aligned}$$

The critical elements of both post- and pre-money investment value have already been discussed, and the computations are straightforward. The importance of these terms to the financial analyst derives from their common usage among VC investors. The pre- and post-money investment values represent estimates, albeit crude ones, of the value of the firm's equity *on the date of* the financing.

Refining the Deal Structure

Entrepreneurs are often surprised and frustrated to learn how much of the firm they must give up to obtain VC financing. In this section, we discuss how the entrepreneur might lower the rate of return required by the venture capitalist by assuming more of the risk of the venture. Although we cannot explicitly describe how this trade-off works, we can describe how the structure of the deal can be adjusted to reallocate risk and return between the entrepreneur and the private equity investor. We consider two alternative ways to structure the venture capital investment. These include the use of *staged-financing* commitments and issuing a different type of security (e.g., preferred stock as opposed to common stock).

Using Staged-Financing Commitments

As we mentioned earlier, the rates of return required by venture capitalists are quite high because the investments are quite risky and the cash flow forecasts tend to be overly optimistic. Thus, anything that can be done to lower the venture capitalist's risks and signal the entrepreneur's confidence will result in lower required rates of return from the VC.

Because venture capitalists tend to be skeptical about cash flow forecasts and want to limit their risks, they generally make what is known as *staged-financing commitments*. Instead of providing the entrepreneur with all the required funding up front, the VC generally makes partial or staged investments as the firm meets predetermined performance milestones and as money is needed. This process gives the VC control over the firm's access to capital and thereby reduces the risk of the VC's investment. If the VC is not satisfied with the firm's progress, then the next installment is simply not made.¹⁰

Let's return to our Bear-Builders.com example to illustrate how staged financing can reduce the cost of financing to the entrepreneurial firm. In our original example, the entrepreneur needed \$2 million, and the venture capitalist was willing to provide the funds in return for an annual rate of return of 50%. Thus, the venture capitalist required that the value of its invested capital be worth $\$15,187,500 = \$2,000,000 (1 + .50)^5$ in five years. Because the firm's equity is estimated to be worth \$27,300,000 at this time, we calculated that the venture capitalist must receive $\$15,187,500 / \$27,300,000 = 56\%$ of the equity of the firm.

Now let's consider a staged-investment alternative in which the venture capitalist invests \$1 million initially (still requiring a 50% rate of return, because this is first-stage

¹⁰ We will have more to say about staged financing later, when we discuss options.

capital) but invests the second \$1 million two years later, on the condition that the firm has achieved certain performance benchmarks. Because the second-stage investment is less risky than the first, we assume that the VC requires only a 30% return on this second infusion of capital.

Using staged commitments, the venture capitalist requires the following dollar return at the end of five years (assuming the second-stage financing is forthcoming):

Financing Stages	Funding Provided	Return Required at the End of Year 5
First-stage investment	\$1,000,000	$\$1,000,000 (1 + .50)^5 = \$7,593,750$
Second-stage investment	<u>\$1,000,000</u>	<u>$\\$1,000,000 (1 + .30)^3 = \\$2,197,000$</u>
Total	\$2,000,000	\$9,790,750

The VC now requires \$9,790,750 for his or her \$2,000,000 investment, which represents $\$9,790,750/\$27,300,000 = 36\%$ of the value of the firm's equity at the end of five years. This dramatic reduction (from 56% to 36%) represents the impact of two things: First, the VC firm pays only \$1 million in capital for the first two years (not \$2 million as before); second, the VC's required return in the second-stage financing is only 30%.

Note that the above analysis assumes that the second-stage financing is provided with certainty. This, however, is not necessarily the case, because the VC has control over the decision to invest at the second stage and will do so only if the investment looks favorable at that time. In making a staged commitment, the VC has the "option" to invest at the second stage, rather than an obligation, and will choose not to invest more money if the firm does not perform as planned. Because this option is valuable, the venture capitalist is willing to provide financing at more favorable terms.

It should be stressed that the entrepreneur gives up less of his or her business, but there is no free lunch here. The entrepreneur is required to take on more risk. If the performance benchmarks are not met after two years, he or she will not be able to obtain the necessary funds to remain in business. But if the entrepreneur is very confident about meeting the benchmark, then the staged financing may be preferable. Given this, a lack of willingness to accept a staged-financing commitment by an entrepreneur would send a negative signal to the VC.

Using Debt or Preferred Stock

A confident entrepreneur may also get more-favorable financing by assuming more of the risk of the venture and limiting the VC's risk exposure by issuing debt or preferred equity securities. The VC's risk is reduced by virtue of the superior claim that debt and preferred stock enjoy over common stock. Again, there is no free lunch here; the entrepreneur receives less-costly financing by limiting the VC's risk exposure and in the process bears more of the risk himself or herself.

To demonstrate how this works, we use the Bear-Builders.com example again. This time, instead of issuing common stock to the venture capitalist, the firm issues convertible preferred stock with a dividend rate of 8%, or an annual dividend of \$160,000 ($\$2 \text{ million} \times .08 = \$160,000$). The preferred stock can be converted into the firm's common stock at the discretion of the VC. (There is generally no incentive on the part of the investor to convert early.) If the VC wants a 40% return and the investment horizon is five years, the cash flow stream to the VC consists of the following:

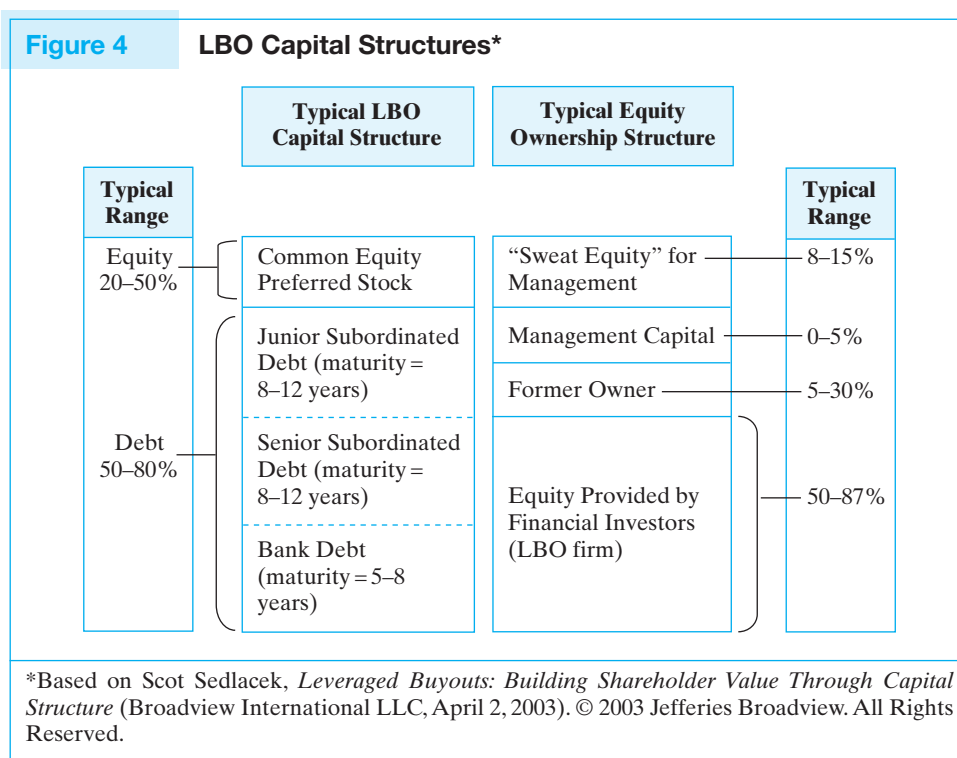
Valuation in a Private Equity Setting

VC's hoped-for rate of return	40%					
VC's investment horizon	5					
Annual interest/Dividend payments	\$ 160,000					
Financing provided by the VC	\$ (2,000,000)					
	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Dividends	\$ (2,000,000)	\$ 160,000	\$ 160,000	\$ 160,000	\$ 160,000	\$ 160,000
Conversion value in five years						\$ 9,003,386
VC cash flows	\$ (2,000,000)	\$ 160,000	\$ 160,000	\$ 160,000	\$ 160,000	\$ 9,163,386

The key calculation we need to make is the required conversion value of the shares that will be realized in Year 5. It turns out that if the preferred shares can be converted into \$9,003,386 worth of common stock, then the VC's \$2 million investment will realize the VC's desired 40% annual rate of return. This, in turn, requires that the VC be given 32.98% of the firm's shares that were earlier valued at \$27,300,000—a substantial savings over the 56% ownership share that was required when using common stock.

4 VALUING LBO INVESTMENTS

Leveraged buyouts, or LBOs, represent a business acquisition strategy whereby an investor group acquires all the equity of a firm and assumes its debts.¹¹ The acquisition is financed predominantly with debt—hence the term “leveraged” buyout. Figure 4 illustrates the



¹¹ When the acquisition is made in conjunction with the firm's management, we refer to the transaction as a *management buyout* or *MBO*.

PRACTITIONER INSIGHT

How Private Equity Firms Create Value— An Interview with Jack Furst*

Over the last 25 years, we've seen seismic shifts in the way private equity investors generate investment returns. At the early part of this decade, we retooled our investment approach to account for changes in lending levels and now seek to add value to our investments using three primary tools.

First, we follow a time-tested/proven investment strategy. We invest in undermanaged, underappreciated, or undercapitalized assets that can be transformed into much-larger, more-valuable companies by investing new capital and creating well-defined operating strategies. This investment approach allows us to generate value for investors by taking advantage of the price premium paid for large public companies compared to small, privately held companies. This strategy can be further simplified by focusing on specific sectors. HM Capital, for example, primarily invests in the energy, food, and media industries—sectors in which the firm has a long history of success and a broad network of contacts.

A recent example of this investment strategy is HM Capital's investment in Regency Energy Partners in December 2004. Prior to acquisition, Regency was a set of disparate natural gas gathering and processing assets in the Southwest. In just 14 months,

HM Capital transformed Regency from a group of natural gas assets to a significant industry player and took the company public. The IPO provided HM Capital with a means of returning all invested capital to investors, in a structure that allowed our investors to maintain a large stake in Regency.

The second tool we use is *Change Capital*. We transform our acquisitions through strategies designed to grow revenues, increase profits, and optimize assets. At HM Capital, we refer to these strategies and the activist role we play in transforming our investments as *Change Capital*. We often acquire companies that have not been run as efficiently as we think is possible. This is sometimes due to the fact that the company is a division of a larger company that has been ignored by the management of the parent company. In other instances, the acquired company is being run by a management team that has little ownership in the company and minimal incentive to manage the company in a lean fashion. In either instance, our goal is to align management's interests with ours to reduce operating costs and create value for owners.

During our review of an investment opportunity, we identify the changes that will need to be implemented post-acquisition and whether the right

proportion of the total financing that comes from debt and how it varies over time with market conditions and typically constitutes 50–80% of the capital structure of the LBO.

The extreme use of financial leverage is a defining characteristic of LBOs when compared to firms' pre-LBO capital structures. Why is it that private equity investments are so highly levered? Perhaps private equity sponsors can more easily attract investors by providing highly levered returns rather than more-modest, unlevered returns. However, this "pure leverage" effect cannot create value on its own. Financial leverage also has an important bonding effect: By saddling their investments with a substantial amount of debt, private equity sponsors force managers to make the tough choices that may be necessary in turnaround situations. In addition, the fact that sophisticated financial institutions are willing to provide the debt for these highly levered entities should provide some comfort for the equity investors.

management team is in place to execute these strategies. We continue to work with management post-acquisition to further refine and implement our *Change Capital* strategies.

Going back to the Regency example, HM Capital's investment thesis was that the company was an underutilized midstream asset with attractive internal growth characteristics. Along with Regency management, HM Capital developed a *Change Capital* plan to invest \$140 million to expand Regency's transmission capacity and transform it from a natural gas gathering- and processing-focused business to a natural gas-transmission business. This capital expenditure quadrupled Regency's pipeline capacity, doubled its cash flow, and helped position the company for an IPO.

The third tool at our disposal is the use of financial leverage as we engineer our capital structure. Much has been made over the amount of debt utilized by private equity firms. In fact, we were typically referred to as leveraged buyout (LBO) investors during the 1980s because we used high levels of debt to fund our acquisitions. Today, we utilize leverage to provide us with the financial flexibility needed to implement our *Change Capital* strategies and generate a higher rate of return on our investments. When used properly, financial leverage is simply an equity-return accelerant that turns doubles into triples and triples into home runs.

The private equity industry has changed significantly because I first started in the business. Leverage is more optimally used to balance the debt and equity requirements of the capital structure, and the size of transactions has grown dramatically. It is not uncommon today to have private equity firms join forces to acquire public companies for \$15 billion to \$25 billion. In addition, the objective of today's private equity investor has evolved—take a company private, restructure its cost structure, and then take the firm back public or sell it to a private investor. Thus, the approach to making money has shifted from “break-ups” in the 1970s and 1980s to “buy and build” in the mid-1980s through the 1990s to “buy, restructure, and build.” The more things change, the more they stay the same. Private equity as an asset class is as attractive as it has ever been.

*Jack D. Furst is an Adjunct Professor of Finance at the University of North Texas. He co-founded Hicks, Muse, Tate & Furst in 1989, renamed HM Capital Partners in 2006. Mr. Furst is a philanthropist and is passionate about the mission of the Boy Scouts of America. He received his BS from the Carey School of Business at Arizona State University and his MBA from the McCombs School of Business at the University of Texas.

Alternative LBO Acquisition Strategies—Bust-Ups and Build-Ups

As the Practitioner Insight box above describes, the use of the LBO acquisition strategy was very popular in the 1980s. Most buyouts in that era were known as “bust-up” LBOs. The term *bust-up* refers to the fact that once control of the acquired company is complete, the new owner sells off some (or all) of the firm's assets and uses the proceeds to repay the debt used to finance the acquisition. These acquisitions are also associated with efforts to increase efficiency in the operations of the remaining assets, which often entails reductions in the firm's workforce.

Beginning in the 1990s, a different private equity strategy evolved that creates a large public company from the acquisition of an initial company (the “platform company”), followed by a series of smaller add-on acquisitions. This strategy, described in the Practitioner Insight box interview with Jack Furst above, is generally referred to as the “build-up” strategy.

INDUSTRY INSIGHT
Leveraged Buyouts—Bust-Ups to Build-Ups

In the 1980s, the term *leveraged buyout*, or simply *LBO*, became synonymous with corporate greed. Books and movies such as *Wall Street*, *Barbarians at the Gate*, and *Other People's Money* glamorized the process of buying a company using borrowed money to finance most of the purchase. Technically, an LBO is simply the acquisition of a company using lots of debt and very little equity. However, the extreme use of debt financing has important behavioral implications. The principal and interest requirements of the debt are usually so onerous as to effectively bond the acquired firm's management to a strict regime of maximizing the cash flow potential of the firm. This can be a very important source of value for a firm that was not efficiently run prior to the LBO.

A popular form of LBO in the 1980s involved acquiring a target firm that was not performing up to its potential. The acquiring firm would then sell off excess capacity (productive assets) and lay off workers at the target firm to make the remaining assets more productive. This type of LBO became known as the “bust-up LBO.”

In many cases, in a matter of years or sometimes only months, the LBO firm that made the acquisition realized a huge profit from the resale of the acquired firm. Most of the purchase price was financed by borrowing with very high-credit-risk *junk bonds*, and the interest and principal payments were made using the acquisition's future cash flows (a combination of the profits of the firm and the sale of assets in the case of “bust-up” LBOs). Thus, in a sense, the acquired company essentially paid for itself, while the managers of the LBO firm captured any increase in value of the firm as their profit.

More recently, the “build-up LBO” has been utilized to consolidate firms in fragmented industries with large numbers of competitors and the potential for economies of size and scale of operations. The process entails first acquiring a “platform,” or base company, and then purchasing a number of related firms, so as to build a core firm of sufficient size to be sold in the public markets.

Example—Build-Up LBOs

The build-up LBO is a special case of a roll-up merger strategy, where an acquiring firm purchases a group of firms in the same or very closely related businesses and then combines them into a single entity. For example, Hicks, Muse, Tate & Furst—a private equity firm in Dallas, Texas (and predecessor of HM Capital)—purchased Berg Electronics from E I DuPont in 1993 for \$370 million. The acquisition became the platform company to which over \$100 million in acquisitions were added before Berg's initial public offering (IPO) in 1996.¹²

To illustrate the valuation issues involved in a build-up LBO, we consider a typical build-up transaction by a private equity partnership. The principals of our hypothetical private equity firm, Hokie Partners LP, are considering the acquisition of PMG Inc., an electronics firm that manufactures printed circuit boards used in a wide variety of

¹² For additional details, see S. Kaplan, J. Martin, and R. Parrino, Berg Electronics Corporation, <http://ssrn.com/abstract=256107>.

applications. PMG is a wholly owned subsidiary of a large chemical company that has decided that the business is no longer critical to the firm's future success and seeks to sell PMG to the highest bidder.

Acquisition data for PMG found in Table 1 indicate that PMG has current earnings before interest, taxes, depreciation, and amortization expense (EBITDA) of \$100 million. The acquisition is expected to require a purchase price equal to five times the current level of EBITDA, or \$500 million. In addition to the equity investment from Hokie Partners and its limited partners, the acquisition is financed with 75% debt that has an interest rate of 14%. The debt has covenants that require that all excess cash be used to retire principal, which means that the equity investors will not receive anything until Year 5, when the firm is sold. This requirement is commonly referred to as the *cash sweep*.

Hokie Partners believes that it can grow EBITDA at a rate of 10% per year for five years and then sell the firm, hopefully for six times EBITDA. When Hokie sells the firm, the firm's outstanding debt will be repaid and the remaining funds distributed to the equity investors.

Estimating Equity Returns for the Platform Company

In Table 2, we evaluate the equity free cash flows (FCFs) and corresponding compound rate of return for the case where the estimates set out in Table 1 are realized. Recall the definition of Equity FCF, which focuses on the cash flows to the common stockholder (as contrasted with the firm free cash flows).¹³ Because of the cash-sweep requirement for the firm's debt, the Equity FCFs for each of the five years of the planning period are set to zero,¹⁴ which implies that the projected principal payments on the firm's debt

Table 1 PMG Inc. Acquisition Data (\$ millions)

Earnings Estimates		Acquisition and Sale EBITDA Multiples	
Current-year EBITDA (millions)	\$100.00	Purchase multiple—Platform company (PMG Inc.)	5
Growth rate in EBITDA	10%	Purchase multiple—Add-on company (Centex)	3
Planned holding period	5 years	Company sale (harvest) multiple	6
Corporate tax rate	35%		
Depreciable life of assets	10 years		
Depreciation expense (Year 0)	\$ 40.00		
LBO Capital Structure			
Debt/Assets (book values)	75%		
Interest cost	14%		
Annual CAPEX	\$ 50.00		

¹³ Equity FCF = Net income + Depreciation Expense – Capital Expenditures (CAPEX) – Change in Net Working Capital – Principal Payments + New Debt Issued.

¹⁴ Note in Table 2: The Equity FCFs are forced to equal 0 through the retirement of \$3.13 million in debt in Year 1, thereby reducing the firm's debt from \$375 million to \$371.88 million; the retirement of \$12.31 million in Year 2 reduces the debt further to \$359.57 million; and so forth.

Table 2 Estimating Equity Returns from the Acquisition of PMG Inc. (\$ millions)

Panel a. Projected Net Income and Equity FCF						
Pro Forma Income Statements	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
EBITDA		\$ 110.00	\$ 121.00	\$ 133.10	\$ 146.41	\$ 161.05
Less: Depreciation		(45.00)	(50.00)	(55.00)	(60.00)	(65.00)
EBIT		\$ 65.00	\$ 71.00	\$ 78.10	\$ 86.41	\$ 96.05
Less: Interest		(52.50)	(52.06)	(50.34)	(47.11)	(42.14)
Earnings before taxes		\$ (12.50)	\$ 18.94	\$ 27.76	\$ 39.30	\$ 53.91
Less: Texas		(4.38)	(6.63)	(9.72)	(13.75)	(18.87)
Net income		\$ 8.13	\$ 12.31	\$ 18.04	\$ 25.54	\$ 35.04
Calculation of Equity FCF						
Plus: Depreciation		\$ 45.00	\$ 50.00	\$ 55.00	\$ 60.00	\$ 65.00
Less: CAPEX		(50.00)	(50.00)	(50.00)	(50.00)	(50.00)
Less: Principal payments		(3.13)	(12.31)	(23.04)	(35.54)	(50.04)
Equity free cash flow		\$ —	\$ —	\$ —	\$ —	\$ —
	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Outstanding loan	\$375.00	\$ 371.88	\$359.57	\$336.52	\$ 300.98	\$ 250.93
Panel b. Valuing the Platform-Company LBO in Year 5 and the IRR on the Investment						
	Year 0	Year 5				
Debt	\$ 375.00	\$ 250.93				
Equity	\$ 125.00	\$ 715.37				
Firm	\$ 500.00	\$ 966.31				
Internal Rates of Return (IRR)						
Equity (with debt—levered)	41.75%					
Equity (no debt—unlevered)	22.46%					

equal the amount of cash left over after meeting all the firm's required expenditures (including new capital equipment purchases).

Note that the transaction requires an investment of \$500 million, and if all works as planned, it will produce an EBITDA of \$161.05 million in five years. Under this scenario, Hokie Partners will sell PMG for the projected six times EBITDA, which implies a value of \$966.31 million and an equity value of \$715.37 million (net of the outstanding debt in Year 5 of \$250.93 million). Given these projections, the equity investor's original \$125 million investment grows to \$715.37 million in five years, which represents a compound annual rate of return of 41.75%.¹⁵ This, of course, should be viewed as a hoped-for

¹⁵ $\$125 \text{ million}(1 + \text{IRR})^5 = \715.37 million , such that IRR is 41.75%.

(rather than expected) return, based on a scenario where EBITDA grows quite quickly and the firm is sold for a higher multiple than was paid for it. The returns are also quite high because they represent the returns to a highly levered equity investment.

To understand the effects of financial leverage on the equity return, we consider the *unlevered* returns on this investment given the same scenario. If the equity investor put up the full \$500 million acquisition price, then the investment would have thrown off cash flows each year equal to the principal and after-tax interest. Again assuming the firm sells for \$966.31 million after five years, the compound annual rate of return is only 22.46% per year.¹⁶ This rate of return is still high, but much lower than the 41.75% levered-equity rate of return. Similarly, we can consider the returns on this investment under a much less favorable scenario where the EBITDA growth rate is 4% rather than 10% and the EBITDA multiple in Year 5 is only four instead of six times. In this scenario, the equity holders realize a value in five years of only \$156.95 million, which implies a levered return of only 4.66%. In this less-favorable scenario, the unlevered return is 7.90%, which exceeds the levered return because the investment earns less than the interest rate on the debt. Financial leverage is truly a two-edged sword! It magnifies returns when outcomes are favorable, but decreases returns when outcomes are not favorable.

Analyzing the Returns from the Add-On Investment

The second stage of a build-up LBO involves the acquisition of one or more “add-on” investments. To illustrate how to evaluate these add-on acquisitions, we consider the acquisition details for Centex PowerPak Systems Inc., found in Table 3.

This acquisition differs from the platform-company acquisition in two important respects. First, it is a much smaller company, with annual EBITDA of only \$12 million in the current year. As discussed earlier in the Practitioner Insight box featuring Jack Furst, small companies are often acquired for lower price multiples, and we will assume that in this case, Centex can be purchased at three times its current level of EBITDA, or \$36 million. Adding

Table 3 Acquisition Details for Centex PowerPak Systems Inc. (\$ millions)

Current-year EBITDA	\$12.00	EBITDA acquisition multiple	3 times
Growth rate in EBITDA	10%	Planning period	5 years
Corporate tax rate	35%	Annual CAPEX	\$10.00
Depreciable life of assets	10 years	Depreciation expense (Year 0)	\$ 3.00
LBO Capital Structure			
Debt/Assets	100%		
Interest cost (planning period)	14%		
Interest cost (post-planning period)	9%		

¹⁶ The IRR to the equity investor who invests \$500 million in return for the following cash flows (millions) is 22.46%. Note that in Year 5, the investor receives the annual equity cash flow of \$77.43 million plus the anticipated sale value of the firm of \$966.31 million (= \$1,043.74 million).

Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
\$(500.00)	\$37.25	\$46.15	\$55.77	\$66.17	\$1,043.74

the relatively cheap assets of the add-on firm to the more highly valued platform company and selling them for the higher multiple of the platform company in five years provides added value, which private equity firms refer to as *multiples expansion*.

The second difference between the platform-company acquisition and the acquisition of the add-on company relates to financing. The add-on company's debt is generally guaranteed by the platform company, so it is difficult to determine exactly how much of the debt can be attributed to the add-on. In the particular example we are considering, the platform company finances the acquisition entirely with debt.

Once again, the initial Equity FCFs are set to zero because all of the firm's cash flow will be used to pay down its debt. However, the firm's initial cash flows are not sufficient to cover the debt service (interest) and CAPEX requirements (see Table 4), so it will need to borrow more in Years 1 through 4. In Year 1, the cash flows fall short by \$3.3 million, increasing the total debt for the Centex acquisition from \$36 million to \$39.3 million. Beginning in Year 5, the growth in operating income is sufficient to pay interest, cover CAPEX, and pay down the debt of the add-on investment by \$1.45 million.

Table 4 Analysis of the Add-On Acquisition of Centex PowerPak Systems Inc. (\$ millions)

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
EBITDA		\$ 13.20	\$ 14.52	\$ 15.97	\$ 17.57	\$ 19.33
Less: Depreciation		(4.00)	(5.00)	(6.00)	(7.00)	(8.00)
EBIT		\$ 9.20	\$ 9.52	\$ 9.97	\$ 10.57	\$ 11.33
Less: Interest		(5.04)	(5.50)	(5.84)	(6.02)	(6.03)
Earnings before taxes		\$ 4.16	\$ 4.02	\$ 4.14	\$ 4.55	\$ 5.30
Less: Taxes		(1.46)	(1.41)	(1.45)	(1.59)	(1.86)
Net income		\$ 2.70	\$ 2.61	\$ 2.69	\$ 2.96	\$ 3.45
Plus: Depreciation		\$ 4.00	\$ 5.00	\$ 6.00	\$ 7.00	\$ 8.00
Less: CAPEX		(10.00)	(10.00)	(10.00)	(10.00)	(10.00)
Less: Principal payments*		3.30	2.39	1.31	0.04	(1.45)
Equity free cash flows**	\$ —	\$ —	\$ —	\$ —	\$ —	\$ 74.36
	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Debt	\$ 36.00	\$ 39.30	\$ 41.68	\$ 43.00	\$ 43.04	\$ 41.59

*Note that principal payments are subtracted, whereas additional borrowing is added. Thus in Year 1 it is necessary to borrow an added \$3.3 million in order to meet the firm's cash flow needs.

**Because 100% of the purchase price is financed by borrowing, the equity investment in Year 0 is zero. Furthermore, in Year 5, Centex's Equity FCF includes the estimated value of the equity in the firm, based upon the assumption that the firm is sold. Because there is no added equity investment in Year 0, the IRR for the add-on investment is undefined (translation—very large). If only 75% of the acquisition price were borrowed, as was the case for the platform company, the IRR would have been 57.88%, which is still quite high.

Table 4 presents an analysis of the projected earnings and equity cash flow from the Centex PowerPak add-on acquisition. In this case, the rate of return on equity will be determined by how much of the new debt that is issued to acquire the firm can be attributed to Centex PowerPak. If we had assumed 75% debt financing, just like the platform company, the IRR on the \$9 million = $25\% \times \$36$ million investment in the add-on company would have been 57.88%, which is still very high.

Does the add-on make sense? If the platform company and the add-on companies are of like risk, and the platform company is deemed a good investment with its 41.75% rate of return on equity, the add-on with its much higher return is an even more attractive investment.

In Panel a of Table 5 (p. 380), we see that the combined (platform plus add-on acquisitions) EBITDA reaches \$180.38 million by Year 5, which, based on the assumed harvest or sale multiple of six times EBITDA at the end of Year 5, implies an expected exit value of \$1,082.26 million. Netting out the \$292.53 million in debt still owed by the combined firms at the end of Year 5 generates an estimated equity value of \$789.74 million. Comparing our initial equity investment of \$125 million to this terminal value, we estimate a compound annual rate of return on the equity investment of 44.58%. In contrast, if no debt had been used and the combined purchase price of \$536 million had been raised through equity financing, the combined Equity FCFs for the platform and add-on investment would have been the following:

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
EBITDA		\$ 123.20	\$ 135.52	\$ 149.07	\$ 163.98	\$ 180.38
Less: Depreciation expense		(49.00)	(55.00)	(61.00)	(67.00)	(73.00)
EBIT		\$ 74.20	\$ 80.52	\$ 88.07	\$ 96.98	\$ 107.38
Less: Interest expense		—	—	—	—	—
EBT		\$ 74.20	\$ 80.52	\$ 88.07	\$ 96.98	\$ 107.38
Less: Taxes		(25.97)	(28.18)	(30.83)	(33.94)	(37.58)
Net income		\$ 48.23	\$ 52.34	\$ 57.25	\$ 63.04	\$ 69.80
Plus: Depreciation		49.00	55.00	61.00	67.00	73.00
Less: CAPEX		(60.00)	(60.00)	(60.00)	(60.00)	(60.00)
Less: Principal payments		—	—	—	—	—
Equity FCF	\$ (536.00)	\$ 37.23	\$ 47.34	\$ 58.25	\$ 70.04	\$ 1,165.06

The compound annual rate of return to the equity shareholders based on the Equity FCFs in the bottom row of the above table, then, is only 23.07%, which is quite high for an unlevered return.

Where is the value creation from this strategy coming from? There are two possible sources of synergies that need to be considered in this strategy that can create what private equity professionals refer to as “multiples expansion.” The first is the fact that the combined company (build-up LBO) will be larger and more liquid than the platform company or the individual add-ons it acquires. Consequently, the combined firm may sell for a higher multiple when it is sold in the public market. The second is that the

Table 5 Analysis of the Build-Up LBO (\$ millions)**Panel a. Pro Forma Income Statements and Equity FCF**

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
EBITDA		\$ 123.20	\$ 135.52	\$ 149.07	\$ 163.98	\$ 180.38
Less: Depreciation expense		(49.00)	(55.00)	(61.00)	(67.00)	(73.00)
EBIT		\$ 74.20	\$ 80.52	\$ 88.07	\$ 96.98	\$ 107.38
Less: Interest expense		(57.54)	(57.56)	(56.17)	(53.13)	(48.16)
EBT		\$ 16.66	\$ 22.96	\$ 31.90	\$ 43.85	\$ 59.21
Less: Taxes		(5.83)	(8.03)	(11.16)	(15.35)	(20.73)
Net income		\$ 10.83	\$ 14.92	\$ 20.73	\$ 28.50	\$ 38.49
Plus: Depreciation		49.00	55.00	61.00	67.00	73.00
Less: CAPEX		(60.00)	(60.00)	(60.00)	(60.00)	(60.00)
Less: Principal payments*		0.17	(9.92)	(21.73)	(35.50)	(51.49)
Equity FCF**	\$ (125.00)	\$ —	\$ —	\$ —	\$ —	\$ —
	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Debt	\$ 411.00	\$ 411.17	\$ 401.25	\$ 379.52	\$ 344.02	\$ 292.53

Panel b. Valuing the Build-Up (Combined) LBO in Year 5 and the IRR on the Investment**Evaluation of IRR on the Platform-Company Investment**

	Year 0	Year 5
Debt	\$ 411.00	\$ 292.53
Equity	\$ 125.00	\$ 789.74***
Firm	\$ 536.00	\$ 1,082.26

Internal Rates of Return (IRR)

Equity with debt (levered)	44.58%
Equity with no debt (unlevered)	23.07%

*Note that principal payments are subtracted, whereas additional borrowing is added. Thus in Year 1 it is necessary to borrow an added \$3.3 million in order to meet the firm's cash flow needs.

**Because 75% of the purchase price for the platform company and 100% of the purchase price of the add-on are financed by borrowing, the equity investment in Year 0 is \$125 million. Furthermore, in Year 5, Centex's Equity FCF includes the estimated value of the equity in the two acquisitions, based upon the assumption that the firm is sold.

*** Equity Value (Year 5) = Firm Value (Year 5) – Interest-Bearing Debt (Year 5) + Cash (Year 5)

thus,

Equity Value (Year 5) = \$1,082.26 – \$292.53 = \$789.74

combined company is more diversified and will thus have a higher debt capacity, which can also result in a higher multiple.

A Limitation of the Industry Private Equity (LBO) Valuation Approach

A weakness of the private equity valuation approach, illustrated in Figure 5, is that there are no direct comparisons of the risks associated with these investments. For example, the risk of the build-up acquisition strategy is an important determinant of whether the 44.58% anticipated return is sufficient to warrant engaging in the investment strategy.

This does not imply that the managers of private equity firms ignore risk and simply choose the investments with the highest projected rates of return. Indeed, it would not be unusual for a private equity firm to select an investment with a projected rate of return of 22% over an alternative investment with a projected rate of return of 28% if the latter were deemed to be much more risky. However, the point remains that the industry-based LBO valuation approach that we describe in this chapter does not explicitly account for risk.

Explicitly accounting for risk in these transactions is somewhat difficult because of the changing use of financial leverage over the life of the investment. The transactions are very highly levered at the outset, but due to the cash-sweep provision typically

Figure 5

Summary of the Industry Private Equity Valuation Method

Step 1: Estimate EBITDA at the end of the planned holding period by preparing pro forma income statements such as those found in Tables 4 and 5. This entails a careful analysis of the acquired firm's past performance and the planned changes the private equity firm has in mind.

Step 2: Estimate the enterprise value at the end of the planned holding period as the product of the predicted EBITDA for Year N and an appropriate EBITDA valuation multiple, $M_{\text{EBITDA Year } N}$.

$$\text{Enterprise Value}_{\text{Year } N} = M_{\text{EBITDA Year } N} \times \text{EBITDA}_{\text{Year } N} \quad (4)$$

Step 3: Calculate the value of the equity of the acquired firm at the end of the planned holding period as follows:

$$\begin{aligned} \text{Equity Value}_{\text{Year } N} = & \text{Enterprise Value}_{\text{Year } N} \\ & - \text{Interest-Bearing Debt}_{\text{Year } N} + \text{Cash}_{\text{Year } N} \end{aligned} \quad (5)$$

We calculate the firm's outstanding interest-bearing debt at the end of the planned holding period (Year N) by subtracting all the principal payments made during the holding period from the initial debt used to finance the acquisition, plus any added borrowing up until Year N . Cash is added back, under the assumption that it is not accounted for in the estimation of enterprise value and constitutes a nonoperating asset.

Step 4: Calculate the IRR of the equity investment as follows:

$$\text{Equity Value}_{\text{Year } 0} = \frac{\text{Equity Value}_{\text{Year } N}}{(1 + \text{IRR})^N} \quad (6)$$

contained in the debt covenants, the debt ratio and the risk of the project decline over time (see Panel a of Table 5). The adjusted present value (APV) model provides an appropriate method to value businesses whose capital structures are not stationary over time but change in a predictable fashion. We consider this approach in the next section.

Valuing PMG Inc. Using the Hybrid APV Approach

This section applies the APV approach (reviewed in Figure 6) to value PMG Inc. To implement this approach, we use the cash flow forecasts from Panel a of Table 6 for a five-year planning period (N), plus an estimate of the terminal value for the enterprise in Year N . Panel a of Table 6 provides estimates for both the unlevered operating cash flows (Firm FCFs) and interest tax savings for this acquisition. The Firm FCFs grow

Figure 6 Review of the Hybrid Adjusted Present Value (APV) Model

Enterprise value is defined as follows:

$$\text{Enterprise Value (APV)} = \text{Present Value} \left(\begin{array}{c} \text{Planning} \\ \text{Period} \\ \text{Cash Flows} \end{array} \right) + \text{Present Value} \left(\begin{array}{c} \text{Post-Planning-} \\ \text{Period} \\ \text{Cash Flows} \end{array} \right) \quad (7)$$

where

$$\text{Present Value} \left(\begin{array}{c} \text{Planning} \\ \text{Period} \\ \text{Cash Flows} \end{array} \right) = \sum_{t=1}^N \frac{\text{Firm FCF}_t}{(1 + k_u)^t} + \sum_{t=1}^N \frac{(\text{Interest}_t \times \text{Tax Rate})}{(1 + k_d)^t} \quad (8)$$

and

$$\text{Present Value} \left(\begin{array}{c} \text{Post-Planning-} \\ \text{Period} \\ \text{Cash Flows} \end{array} \right) = \frac{\text{Terminal (Enterprise) Value}}{(1 + k_u)^N} \quad (9)$$

Legend

Firm FCF = Firm free cash flow, which is the same as the unlevered equity free cash flow because its calculation does not entail deductions for either principal or interest. When Firm FCF is discounted to the present at the unlevered cost of equity capital (k_u), it represents the value of the unlevered equity of the firm.

Interest Expense_{*t*} × Tax Rate = Interest tax savings, which represent the reduction in taxes owed that results from interest expense deductions.

Terminal (Enterprise) Value = Enterprise value estimated at the end of the planning period, which represents the present value of the post-planning-period Firm FCFs. The terminal value is estimated as a multiple of earnings before interest, taxes, depreciation, and amortization (EBITDA) for year N (i.e., $M_{\text{EBITDA}} \times \text{EBITDA}_N$).

N = Planning period for which detailed estimates of cash flows are prepared (usually three to five years).

k_u = Unlevered cost of equity capital.

k_d = Cost of debt financing.

Table 6 APV Valuation of the PMG Inc. Build-up LBO Transaction

Panel a. Projected Free Cash Flow (FCF) and Interest Tax Savings						
	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
EBITDA		\$ 123.20	\$ 135.52	\$ 149.07	\$ 163.98	\$ 180.38
Less: Depreciation		(49.00)	(55.00)	(61.00)	(67.00)	(73.00)
EBIT		\$ 74.20	\$ 80.52	\$ 88.07	\$ 96.98	\$ 107.38
Less: Taxes		(25.97)	(28.18)	(30.83)	(33.94)	(37.58)
NOPAT		\$ 48.23	\$ 52.34	\$ 57.25	\$ 63.04	\$ 69.80
Plus: Depreciation		49.00	55.00	61.00	67.00	73.00
Less: CAPEX		(60.00)	(60.00)	(60.00)	(60.00)	(60.00)
Firm FCF		\$ 37.23	\$ 47.34	\$ 58.25	\$ 70.04	\$ 82.80
Interest expense		\$ 57.54	\$ 57.56	\$ 56.17	\$ 53.13	\$ 48.16
Interest tax savings		\$ 20.14	\$ 20.15	\$ 19.66	\$ 18.60	\$ 16.86
Debt	\$ 411.00	\$ 411.17	\$ 401.25	\$ 379.52	\$ 344.02	\$ 292.53
Panel b. APV Valuation Analysis						
Assumptions:						
Terminal value multiple	6 times EBITDA _{Years 5}					
Interest rate on debt	14%					
Unlevered beta	1.19					
Risk-free rate	5.00%					
Market equity risk premium	5.50%					
Unlevered cost of equity	11.56%					
Tax rate	35%					
Hybrid APV Valuation (\$ millions):						
	Present Value of					
	Unlevered Equity Free Cash Flows (discount rate = $k_{\text{Unlevered}}$)		Interest Tax Savings (discount rate = k_d)	Total		
Planning period cash flows	\$206.48		\$66.20	\$ 272.69		
Terminal-value cash flow (discount rate = k_u)				\$ 626.28		
Enterprise value (Year 0)				\$ 898.97		
Less: Debt (Year 0)				(411.00)		
Equals: Equity value (Year 0)				\$ 487.97		

steadily over the planning period (N), from \$37.23 million to \$82.80 million, as a result of the planned growth in revenues. On the other hand, the interest tax savings decline from \$20.14 million to \$16.86 million as a result of the cash-sweep provision in the firm's debt agreement, which requires that the firm pay down the debt using all available cash flows.

In addition to the cash flows, we need estimates of three rates that we will use to discount the cash flows: First, we will need the firm's borrowing rate (k_d), which is used to discount the interest tax savings. Second, we will need the unlevered cost of equity, k_u , to discount the unlevered equity cash flows over the five-year planning period. Third, we need a cost of capital to discount the terminal value at the end of Year 5, which we will assume is also equal to the cost of equity. The borrowing rate we use to discount the interest tax savings is 14% (Panel b of Table 6). Finally, the unlevered cost of equity is 11.56%. Although the details are not shown here, we use the same procedure to estimate the unlevered cost of equity.

We can now estimate the value of PMG Inc. *as of today* (not at the end of Year 5). Panel b of Table 6 contains the APV estimate of the value of PMG Inc. The terminal value contributes \$626.28 million to the total estimated enterprise valuation of \$898.97 million. Netting out the firm's initial debt of \$411.00 million produces an estimated value of the private equity firm's equity investment of \$487.97 million.

Is the proposed acquisition worthwhile? These numbers indicate that this is a great investment because it generates an equity value of \$487.97 million on an equity investment of only \$125 million. However, it should be emphasized that the estimated cash flows used in our analysis may correspond more closely to hoped-for cash flows than to expected cash flows, which suggests that one might want to use a much higher discount rate to value them. Consequently, before undertaking an investment of this magnitude, the private equity firm would be wise to engage in a wide range of sensitivity analyses of the key value drivers of the investment.

Private Equity Value Drivers

There are five drivers of value in the example just discussed. These are: (1) the estimate of the first year's EBITDA, (2) the anticipated rate of growth in EBITDA for each year of the planning horizon, (3) the level of capital expenditures required each year, (4) the EBITDA multiple used to estimate the terminal value in five years, and (5) the unlevered cost of capital. We will look at the sensitivity of the project outcome to each of these value drivers.

The simplest form of sensitivity analysis involves calculating the critical values of each of the five value drivers to determine the level of each required to produce an equity value for the investment equal to the \$125 million investment. The results are as follows:

Value Driver (Variable)	Estimated Value	Critical Value	Margin of Error (Estimate – Critical)
Year 1 EBITDA	\$123.20 = \$110 + 13.20	\$76.00	\$47.20
EBITDA growth rate	10%	–3.65%	13.65%
CAPEX	\$60 million/year	\$181.32 million/year	\$121.32 million/year
Harvest multiple	6 times EBITDA	2.523 times EBITDA	3.477 times EBITDA
Unlevered cost of equity	11.56%	26.81%	15.25%

Clearly, this investment creates value even in the face of very significant changes in the individual estimates of the key value drivers. For example, the initial year's projected EBITDA can drop from the projected \$123.2 million to only \$76 million, and the investment will still produce a value of \$125 million.¹⁷ Moreover, the growth rate in EBITDA (holding the other three value drivers at their prior levels) can drop to -3.65% before the value of the investment will drop below the \$125 million in equity capital that is invested in the project. Capital expenditures (CAPEX) can increase to \$181.32 million over the expected \$60 million before the project NPV becomes negative. Similarly, the EBITDA (harvest) multiple at the end of Year 5 that is used to determine the terminal value of the firm can drop to 2.523 times, and the unlevered cost of equity could be as high as 26.81% before the project's value drops below the \$125 million invested in the equity of the deal.

In the preceding analysis, we analyzed only one value driver at a time. Such single-variable analysis can easily understate the true risks of the investment in cases where the value drivers vary together. For example, if both the initial EBITDA estimate and the growth rate in EBITDA are 20% higher than anticipated, and the CAPEX estimate is 20% lower than expected, then an exit multiple of EBITDA of 3.31 times generates an enterprise value just large enough to cover the debt payment, thereby leaving nothing for the equity holders. Although the above scenario may be unlikely, it illustrates the fact that this is a risky investment.

5 SUMMARY

Private equity firms are becoming an increasingly important source of capital for start-up firms as well as for more mature firms that are going through some sort of transition. In most cases, private equity firms evaluate their investments using a variant of the hybrid valuation approach. Typically, the focus of the valuation is on the investor's equity investment, and because there are generally no planning period cash flows to the equity holders, the focus of the valuation is on the firm's terminal value, which is usually calculated as a multiple of EBITDA.

Another notable feature of the valuation approach used by private equity firms relates to their use of discount rates that are much higher than the rates considered in the previous chapters. These investments should have high expected rates of return, given their high risk and illiquidity. However, the discount rates required by private equity firms are high, even given the nature of these investments, and probably reflect the fact that they are used to discount cash flows that are likely to be overly optimistic.

We would again like to emphasize that there are reasons why, in practice, investors often evaluate optimistic or hoped-for cash flows. Because of this, the valuation approaches used in practice often differ from those recommended by academics. We recommend that, in situations where private equity investors have estimates of expected cash flows rather than hoped-for cash flows, they evaluate the investment using the APV approach. Although this approach is not widely used in the private equity industry, we think it is appropriate, given that most of these transactions involve a substantial amount of initial debt that is paid down fairly quickly over the planning period.

In closing, we should point out that this chapter has considered a couple of investments that should really be viewed as options. First, in the venture capital example, we consider staged financing, which involves an initial investment that gives the venture

¹⁷ Note that in this analysis, we assume that both the platform and add-on EBITDAs for Year 1 are changed simultaneously by the sensitivity parameter.

capitalist an option, not an obligation, to make additional investments in the firm when those investments look favorable. Second, in the LBO example, we discussed the investment in a platform company that provides the option—again, not the obligation—to acquire add-on companies in situations where doing so is favorable to the private equity firm. Over the past 20 years, financial economists have developed methods for explicitly valuing investments that include such options.

PROBLEMS

1 USING DEBT FINANCING TO ENHANCE EQUITY RETURNS Kimble Electronics is a small toy manufacturing company with total assets of \$1.5 million. The company has the opportunity to do a leveraged recapitalization that would involve borrowing 30% to 50% of the firm's assets at a rate of 9% per year. Moreover, the firm's annual return on its total investment in assets varies from 4% to 15%.

Analyze the effects of financial leverage on the rate of return earned on Kimble's equity if the firm borrows 30% or 50% of its assets (i.e., complete the following table). (You may ignore taxes in your calculations.)

	15% Return on Assets	4% Return on Assets
30% debt to assets	?	?
50% debt to assets	?	?

2 PRE- AND POST-MONEY VALUATION SPC Venture Partners has evaluated an early-stage investment in a start-up company (Startco). Startco needs \$4 million in financing, and SPC has offered to provide the financing in exchange for a 40% interest in the company.

- What is SPC's post-money valuation of Startco?
- What is the pre-money valuation of Startco implicit in the deal terms offered by SPC Venture Partners?

3 DEAL STRUCTURING CP Venture Partners is looking at the possible investment of \$1 million in an early-stage company (Startco). Given the stage of the investment, CP requires a 40% annual rate of return. The company earned \$750,000 in EBITDA last year, and this amount is expected to grow at a rate of 30% per year over the next five years. Companies such as the subject firm are currently being valued at five times EBITDA, and CP Venture Partners thinks this is a reasonable multiple for the valuation of the firm in five years.

- Assuming that Startco currently has no debt and does not plan to borrow, what do you estimate the enterprise value of the company will be at the end of five years?
- What fraction of Startco does CP Venture Partners need to own at the end of five years in order to realize the required rate of return on its investment?

4 VC VALUATION AND DEAL STRUCTURING Chariot.com needs \$500,000 in venture capital to bring a new Internet messaging service to market. The firm's management has approached Route 128 Ventures, a venture capital firm located in the high-tech start-up

mecca, known as Route 128, in Boston, Massachusetts, which has expressed an interest in the investment opportunity.

Chariot.com's management made the following EBITDA forecasts for the firm, spanning the next five years:

Year	EBITDA
1	−\$175,000
2	75,000
3	300,000
4	650,000
5	1,050,000

Route 128 Ventures believes that the firm will sell for six times EBITDA in the fifth year of its operations and that the firm will have \$1.2 million in debt at that time, including \$1 million in interest-bearing debt. Finally, Chariot.com's management anticipates having a \$200,000 cash balance in five years.

The venture capitalist is considering three ways of structuring the financing:

1. Straight common stock, where the investor requires an IRR of 45%.
2. Convertible debt paying 10% interest. Given the change from common stock to debt, the investor would lower the required IRR to 35%.
3. Redeemable preferred stock with an 8% dividend rate, plus warrants entitling the VC to purchase 40% of the value of the firm's equity for \$100,000 in five years. In addition to the share of the firm's equity, the holder of the redeemable preferred shares will receive 8% dividends for each of the next five years, plus the face value of the preferred stock in Year 5.

TERMINOLOGY

A *convertible* security (debt or preferred stock) is replaced with common stock when it is converted. The principal is not repaid. In contrast, the face value of a security with *warrants* is repaid, and the investor has the right to receive common stock shares by remitting the warrants.

Redeemable preferred stock is typically straight preferred with no conversion privileges. The preferred always carries a negotiated term to maturity, specifying when it must be redeemed by the company (often, the sooner of a public offering or five to eight years). The preferred shareholders typically receive a small dividend (sometimes none), plus the face amount of the preferred issue at redemption, plus a share of the value of the firm in the form of common stock or warrants.

- a. Based on the offering terms for the first alternative (common stock), what fraction of the firm's shares will it have to give up to get the requisite financing?
- b. If the convertible debt alternative is chosen, what fraction of the firm's ownership must be given up?
- c. What rate of return will the firm have to pay for the new funds if the redeemable preferred stock alternative is chosen?
- d. Which alternative would you prefer if you were the management of Chariot.com? Why?

5 VC VALUATION Southwest Ventures is considering an investment in an Austin, Texas-based start-up firm called Creed and Company. Creed and Company is involved in organic gardening and has developed a complete line of organic products for sale to the public that ranges from composted soils to organic pesticides. The company has been around for almost 20 years and has developed a very good reputation in the Austin business community, as well as with the many organic gardeners who live in the area.

Last year, Creed generated earnings before interest, taxes, and depreciation (EBITDA) of \$4 million. The company needs to raise \$5.8 million to finance the acquisition of a similar company called Organic and More that operates in both the Houston and Dallas markets. The acquisition would make it possible for Creed to market its private-label products to a much broader customer base in the major metropolitan areas of Texas. Moreover, Organic and More earned EBITDA of \$1 million in 2009.

The owners of Creed view the acquisition and its funding as a critical element of their business strategy, but they are concerned about how much of the company they will have to give up to a venture capitalist in order to raise the needed funds. Creed hired an experienced financial consultant, in whom they have a great deal of trust, to evaluate the prospects of raising the needed funds. The consultant estimated that the company would be valued at a multiple of five times EBITDA in five years and that Creed would grow the combined EBITDAs of the two companies at a rate of 20% per year over the next five years if the acquisition of Organic and More is completed.

Neither Creed nor its acquisition target, Organic and More, uses debt financing at present. However, the VC has offered to provide the acquisition financing in the form of convertible debt that pays interest at a rate of 8% per year and is due and payable in five years.

- a. What enterprise value do you estimate for Creed (including the planned acquisition) in five years?
- b. If the VC offers to finance the needed funds using convertible debt that pays 8% per year and converts to a share of the company sufficient to provide a 25% rate of return on his investment over the next five years, how much of the firm's equity will he demand?
- c. What fraction of the ownership in Creed would the venture capitalist require if Creed is able to grow its EBITDA by 30% per year (all else remaining the same) and the venture capitalist still requires a 25% rate of return over the next five years?

6 VC VALUATION AND DEAL STRUCTURING SimStar Manufacturing Co. needs \$500,000 to fund its growth opportunities. The founder of the firm has approached Morningstar Ventures (a Phoenix-based venture capital firm), which has expressed an interest in providing the financing if acceptable terms can be worked out.

The venture capitalist asked SimStar's CFO to provide an EBITDA forecast for the next five years. The forecast found below depicts the rapid growth opportunities the firm anticipates:

Year	EBITDA
1	-\$350,000
2	200,000
3	340,000
4	1,050,000
5	1,500,000

If the venture capitalist provides the needed funds, she will plan on an exit after five years, at which time the VC believes that the firm can be sold for six times EBITDA. Moreover, based on pro forma financials and the above projections, the VC estimates that SimStar will have approximately \$1.2 million in debt outstanding, including \$1 million in interest-bearing debt, at the end of the planned five-year investment. Finally, SimStar expects that it will have \$200,000 in cash at the end of five years.

The venture capitalist is considering three ways of structuring the financing:

- a. Straight common stock, where the VC receives no dividends for a period of five years but wants 49% of the firm's shares.
- b. Convertible debt paying 10% interest. Given the change from common stock to debt, the VC wants only 30% of the firm's equity upon conversion in five years.
- c. Redeemable preferred stock with an 8% dividend rate plus warrants for 40% of the firm's equity in five years. Moreover, the exercise costs for the warrants total \$100,000. Note that because this is "redeemable preferred," the investor not only receives an 8% dividend for each of the next five years plus the face value of the preferred stock, but can also purchase 40% of the firm's equity for \$100,000.

Which of the alternatives would you recommend that SimStar's founder select? Explain your decision.

7 VC VALUATION AND DEAL STRUCTURING Brazos Winery was established eight years ago by Anna and Jerry Lutz with the purchase of 200 acres of land. The purchase was followed by a period of intensive planting and development of the grape vineyard. The vineyard is now entering its second year of production.

In March 2009, the Lutzes determined that they needed to raise \$500,000 to purchase equipment to bottle their private-label wines. Unfortunately, they have reached the limits of what their banker can finance and have put all their personal financial resources into the business. In short, they need more equity capital, and they cannot provide it themselves.

Their banker recommended that they contact a venture capital (VC) firm in New Orleans that sometimes makes investments in ventures such as the Brazos Winery. He also recommended that they prepare for the meeting by organizing their financial forecast for the next five years. The banker explained that VCs generally target a five-year term for their investments, so it was important that they provide the information needed to value the winery at the end of five years.

After a careful analysis of their plans for the winery, the Lutzes estimate that earnings before interest, taxes, depreciation, and amortization (EBITDA) in five years will be \$1.2 million. In addition to the EBITDA forecast, the Lutzes estimate that they will need to borrow \$2.4 million by 2013 to fund additional expansion of their operations. Their banker indicated that his bank could be counted on for \$2 million in debt, assuming they were successful in raising the needed equity funds from the VC. Furthermore, the remaining \$400,000 would be in the form of accounts payable. Finally, the Lutzes believe that their cash balance will reach \$300,000 at the end of five years.

The Lutzes are particularly concerned about how much of the firm's ownership they will have to give up in order to entice the VC to invest. The VC offers three alternative ways of funding the winery's \$500,000 financing requirements. Each alternative calls for a different ownership share:

- Straight common stock that pays no dividend. With this option, the VC asks for 60% of the firm's common stock in five years.

- Convertible debt paying 10% annual interest and 40% of the firm's common stock at conversion in Year 5.
- Convertible preferred stock with a 10% annual dividend and the right to convert the preferred stock into 45% ownership of the firm's common stock at the end of Year 5.
 - a. If the VC estimates that the winery should have an enterprise value equal to six to seven times estimated EBITDA in five years, what do you estimate the value of the winery to be in 2013? What will the equity in the firm be worth? (*Hint: Consider both the six- and seven-times-EBITDA multiple.*)
 - b. Based on the deal terms offered, what rate of return does the VC require for each of the three financing alternatives? Which alternative should the Lutzes select based on the expected cost of financing?
 - c. What is the pre- and post-money value of the firm based on the three sets of deal terms offered by the VC? Why are the estimates different for each of the deal structures?
 - d. How is the cost of financing affected by the EBITDA multiple used to determine enterprise value? Is it in the VC's best interest to exaggerate the size of the multiple or to be conservative in his estimates? Is it in the entrepreneurs' best interest to exaggerate the estimated EBITDA levels or to be conservative? If entrepreneurs are naturally optimistic about their firm's prospects, how should the VC incorporate this into his deal-structuring considerations?
 - e. Discuss the pros and cons of the alternative sources of financing.

8 VC VALUATION AND DEAL STRUCTURING In 2008, Dub Tarun founded a firm using \$200,000 of his own money, \$200,000 in senior (bank) debt, and an additional \$100,000 in subordinated debt borrowed from a family friend. The senior debt pays 10% interest, while the sub debt pays 12% interest and is convertible into 10% of the firm's equity ownership at the option of the investor, J Martin Capital. Both debt issues have 10-year maturities. In March 2009, the firm's financial structure appeared as follows:

Dub Tarun Inc., March 2009

Accounts payable	\$100,000
Short-term notes	150,000
Total short-term debt	<u>\$250,000</u>
Senior debt (10% interest rate)	200,000
Sub debt (12% interest rate, convertible into 10% stock)	100,000
Equity (Dub Tarun)	<u>200,000</u>
Total debt and equity	\$750,000

Dub has determined that he needs an additional \$250,000 if he is going to continue to grow his business. To raise the necessary funds, he intends to use an 8% convertible preferred stock issue.

Dub projects that the firm's EBITDA (earnings before interest, taxes, depreciation, and amortization) in five years will be \$650,000. Although Dub isn't interested in selling his firm, his banker recently told him that businesses like his typically sell for five to

seven times their EBITDA. Moreover, by March 2014, Dub expects that the firm will have \$300,000 in cash and that the firm's pro forma debt and equity will be as follows:

Dub Tarun Inc. Pro Forma Financial Structure, March 2014

Accounts payable	\$ 200,000
Short-term notes	250,000
Total short-term debt	\$ 450,000
Senior debt (10%)	400,000
Sub debt (12%, convertible into 10% of the firm's stock)	100,000
Equity (Dub Tarun)	800,000
Additional financing needed	250,000
Total debt and equity	\$2,000,000

- a. What would you estimate the enterprise value of Dub Tarun Inc. to be on March 2014? (*Hint:* Enterprise value is typically estimated for private companies using a multiple of EBITDA plus the firm's cash balance.) If the sub debt converts to common in 2014, what is your estimate of the value of the equity of Dub Tarun in 2014?
- b. If the estimated enterprise value of the firm equals your estimate in Question a, what rate of return does the sub debt holder realize if he converts in 2014? Would you expect the sub debt holder to convert to common stock?
- c. If the new investor were to require a 45% rate of return on his \$250,000 purchase of convertible preferred stock, what share of the company would he need, based on your estimate of the value of the firm's equity in 2014? What is your estimate of the ownership distribution of Dub Tarun's equity in 2014, assuming that the new investor gets what he requires (to earn his 45% required rate of return) and the sub debt holder converts to common? What rates of return do each of the equity holders in the firm expect to realize by 2014, based on your estimate of equity value? Does the plan seem reasonable from the perspective of each of the investors?
- d. What would be Dub Tarun's expected rate of return if the EBITDA multiple were five or seven?
- e. What was the post-investment and pre-investment value of Dub Tarun's equity in 2009, based on the investment of the new investor?

9 LBO VALUATION Randy Dillingwater is the chief investment officer for Clearstone Capital. Clearstone is a private equity firm located in Orlando, Florida, that specializes in what Randy describes as “make-over” or “fixer-upper” investments. The firm tries to find privately held firms whose owners tried to grow their business too fast and ran into liquidity problems. Clearstone has been in this business for 11 years and has had reasonable success.

Clearstone is now completing the investment of its second fund and considering the acquisition of a local manufacturing and distribution company, Flanders Inc. Flanders was founded by Mark Flanders 18 years ago and grew rapidly. Recently, however, the firm made a large acquisition of a competitor firm, and the problems the firm encountered when assimilating the acquisition led to financial difficulties for Flanders. The owner has recently voiced his interest in a buyout proposal to his local banker, who notified Randy (his next-door neighbor) of the opportunity.

Randy contacted Mark, and the two decided to open a dialogue about the possible acquisition of Mark's firm. After several meetings, Mark decided to solicit an offer from Clearstone. In response to Randy's request, Mark supplied him with the following set of pro forma income statements spanning 2010–2014:

Pro Forma Income Statements	2010	2011	2012	2013	2014
EBITDA	\$ 11,000,000.00	\$ 12,100,000.00	\$ 13,310,000.00	\$ 14,641,000.00	\$ 16,105,100.00
Less: Depreciation	(3,900,000.00)	(4,300,000.00)	(4,700,000.00)	(5,100,000.00)	(5,500,000.00)
EBIT	\$ 7,100,000.00	\$ 7,800,000.00	\$ 8,610,000.00	\$ 9,541,000.00	\$ 10,605,100.00
Less: Interest	(6,300,000.00)	(6,235,600.00)	(6,040,288.80)	(5,690,457.10)	(5,159,103.90)
Earnings before taxes	\$ 800,000.00	\$ 1,564,400.00	\$ 2,569,711.20	\$ 3,850,542.90	\$ 5,445,996.10
Less: Taxes	(240,000.00)	(469,320.00)	(770,913.36)	(1,155,162.87)	(1,633,798.83)
Net income	\$ 560,000.00	\$ 1,095,080.00	\$ 1,798,797.84	\$ 2,695,380.03	\$ 3,812,197.27

In addition, Randy asked Mark to estimate capital expenditures for each of the next five years. Mark indicated that he thought the firm would have to spend about \$4 million a year and that the new capital would have a 10-year depreciable life. (Depreciation expense for 2009 was \$3.5 million, such that the addition of \$4 million in capital expenditures will add \$400,000 in added depreciation expense for 2010.)

Mark indicated to Randy that his research suggested that a five-times-EBITDA multiple would be appropriate. Randy, however, was not sure that Clearstone could afford to pay this much for the firm. He decided to do a quick analysis using the LBO method of valuation based on the following assumptions:

- The firm can be purchased for five times the firm's 2009 EBITDA of \$10 million and resold in five years for the same multiple of the firm's Year 5 EBITDA.
- Clearstone will finance 90% of the purchase price using debt that carries a 14% rate of interest. The debt will require a cash sweep, such that all available cash flow will go toward the repayment of the note.
- A tax rate of 30% is assumed in all calculations.
- Capital expenditures (CAPEX) are estimated to be \$4 million per year, and no net new investments in working capital are anticipated.
- Flanders does not carry any excess cash and has no nonoperating assets.
 - a. What is the projected enterprise value of Flanders in five years? What is the estimated value of Clearwater's equity in the firm at the end of five years if everything works out as planned?
 - b. What rate of return should Clearstone expect on its equity in the acquisition under the projections made above?
 - c. After further review, Randy estimated that the firm's operating expenses could be shaved by roughly \$1 million per year. How would this affect your answers to Questions a and b?

10 LBO VALUATION USING THE APV MODEL *This problem utilizes information from Problem 9 but can be analyzed independently.* Randy Dillingwater wanted to perform

a discounted cash flow analysis of the proposed acquisition of Flanders Inc. However, the cash-sweep provision of the debt financing meant that the debt ratio was expected to change systematically over the five-year planned investment period used by Randy's firm (Clearstone Capital). This, Randy recalled, would make the traditional WACC model for valuing firms tough to use. Instead, he decided to use the adjusted present value (APV) model, which separates the valuation of operating and financial cash flows into two separate calculations.

In preparation for his analysis, Randy did some initial research to determine reasonable estimates of the capital market components of the CAPM. First, he noted that the 20-year long-term government bond yield was currently 5% and that the return premium of the equity market in excess of the long-term government bond yield had historically averaged about 5.5%. All that remained to do was estimate the unlevered beta coefficient for Flanders. To perform this analysis, Randy decided that he needed to calculate the unlevered equity betas for a sample of similar firms and use the average of the sample as his estimate of Flanders' unlevered beta. He then collected the following information on four publicly traded firms that were deemed close substitutes for Flanders:

Company	Equity Betas	Debt	Equity Value	Debt to Equity	Marginal Tax Rate
Company A	1.400	\$ 6.58 m	\$ 22.1 m	29.77%	30%
Company B	1.750	11.34 m	25.8 m	43.95%	30%
Company C	1.450	3.42 m	10 m	34.20%	30%
Company D	1.230	6.59 m	25.2 m	26.15%	30%

The equity betas were what Randy referred to as "levered equity betas," for they were calculated using stock market returns, which in turn reflected the firm's use of debt in its capital structure. Consequently, Randy needed to unlever the equity betas. To perform this analysis, Randy decided to assume that the firm's debt had a zero beta, such that the unlevered beta was defined as follows:

$$\text{Unlevered Equity Beta} = \frac{\text{Levered Equity Beta}}{1 + \text{Debt Ratio} (1 - \text{Tax Rate})}$$

- What is your estimate of Flanders' unlevered equity beta? (*Hint: Refer back to Table 1 of section entitled "Estimating a Firm's Cost of Capital".*)
- Using the CAPM, what is the required rate of return that Randy should use to value Flanders' operating cash flows?
- What is the value of Flanders' estimated operating cash flows (based on Mark Flanders' original estimates from Problem 9) for the planning period (2010–2014)? What is the value of the firm's interest tax savings for the same period? (Recall that the firm's borrowing rate from Problem 9 was 14%.)
- Randy decided to estimate the continuing or terminal value of the firm using a five times multiple of the firm's EBITDA. What is the terminal value of the firm today using this procedure, with the terminal value discounted back to 2009 using the unlevered cost of equity capital? What is your estimate of the enterprise value of

Flanders using the APV method (and your estimates from Question c)? What is the estimated value of Clearstone's equity investment based on the APV estimate of enterprise value?

11 LBO VALUATION USING SIMULATION *This problem utilizes information from Problem 9 and requires that the LBO valuation model be constructed.* Randy was happy with the anticipated results from the acquisition of Flanders Inc., but he wanted to do some exploratory analysis of the risks involved in the acquisition so that he might be able to anticipate the risks inherent in the investment. He planned to construct a spreadsheet model using the industry-standard LBO valuation model (using information from Problem 9) and then identify the key sources of risk in the model, constructing what he felt were reasonable characterizations of the probability distributions for each of these variables. Specifically, Randy decided upon the following variables (value drivers) and distributional characterizations:

- Mark Flanders estimated EBITDA growth to be 10% per year, but this rate is far from certain. Randy decided that he would set the most likely growth rate to be 10% but would use a triangular distribution with a minimum value of 0% and a maximum of 15% to characterize the annual rate of growth in EBITDA over the planning period. Moreover, Randy felt that the rate of growth each year would be related to what happened in the subsequent year, so he decided to impose a .80 correlation on the growth rates for successive years. That is, the correlation between the rate of growth in 2012 and 2011 is estimated to be .80, and so forth.
- Another key determinant of the value of the Flanders acquisition is the exit multiple that Clearstone will encounter in five years when it sells the firm. Randy has estimated that they will be able to exit at the same multiple (five) at which they purchase Flanders Inc.; however, this variable is subject to uncertainty. Based upon Clearstone's past experience, Randy estimates that there is a 10% chance that the exit multiple will be as low as three times EBITDA, a 20% chance that it will be four, a 40% chance that it will be five, a 20% chance that it will be six, and a 10% chance that the multiple will be as high as seven.
- a. After incorporating Randy's two assumptions about the uncertainties in the acquisition of Flanders Inc., what is the mean and median rate of return on Clearstone's investment in the firm's equity?
- b. What is the probability that the Flanders acquisition will yield a rate of return to Clearstone that is below the firm's target of 40%?